

Nice to see you!

- Connect to **WiFi** with your Eduroam account or hit me up!
- **Slides and hands-on materials** are accessible on Github:
<https://github.com/Data-Science-Center-UB/Intro-Quantitative-Analysis-R>

Please download the folder **Materials** to your local machine

- You need a local **R/R Studio** installation:
 - R : <https://cran.r-project.org>
 - RStudio: <https://posit.co/download/rstudio-desktop/>



DATA SCIENCE
CENTER



DataNord



University
of Bremen

No Fear of Numbers

Introduction to Quantitative Data Analysis in R

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dsc-ub.de

11 March 2026 ▪ DSC-Training

Schedule For Today

-  09:00 **Welcome and Introduction**
-  09:30 Basic Statistical Concepts
-  09:50 Introduction to R
-  11:20 Univariate Analysis
-  12:30 **Lunch break**
-  13:45 Bivariate Analysis
-  15:15 Multivariate Analysis
-  **16:25 Farewell**

Idea

After an R introduction, we'll go through four practical parts today:

- ❶ Exploring data
- ❷ Univariate analysis
- ❸ Bivariate analysis
- ❹ Multivariate analysis

→ Each part starts with a short input introducing the theory and some basic R concepts.

→ Then you'll have time to work on the exercises independently.

📄 **Exercise sheet (without solutions)** – for those who are already more proficient in R

📄 **Exercise sheet (with solutions and detailed explanations)** – for those who prefer to follow along conceptually

→ After each hands-on block, we'll go through the solutions together and discuss different approaches.

→ Don't worry if you can't finish everything — all materials remain available afterwards.

Housekeeping

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- You need a local R/R Studio installation:
 - R : <https://cran.r-project.org>
 - RStudio: <https://posit.co/download/rstudio-desktop/>
- Please sign the attendance sheet.
- Certificates will be sent after the training.
- Questions welcome anytime 😊

Who are we?

Susanne de Vogel (she/her)

Social Scientist

PhD in Sociology (2019), Martin-Luther-University Halle-Wittenberg

Maryam Movehedifar (she/her)

Statistician

PhD in Statistics (2020), Payame Noor University Tehran

Since May 2024 **Data Scientists** at the **Data Science Center**

- Training and consulting for researchers
- Data science and research data management



Brief round of introductions



- Name (pronouns)
- Institution & discipline
- Previous experience with quantitative data & programming software

Basic Statistical Concepts

The background of the slide is a dark blue gradient. On the right side, there are numerous thin, bright blue lines and dots that appear to be moving or streaming upwards and to the right, creating a sense of dynamic energy or data flow. The overall aesthetic is modern and technological.

Population & Sample

Population is the complete set of individuals that share common characteristics and are the **focus of a quantitative study**.

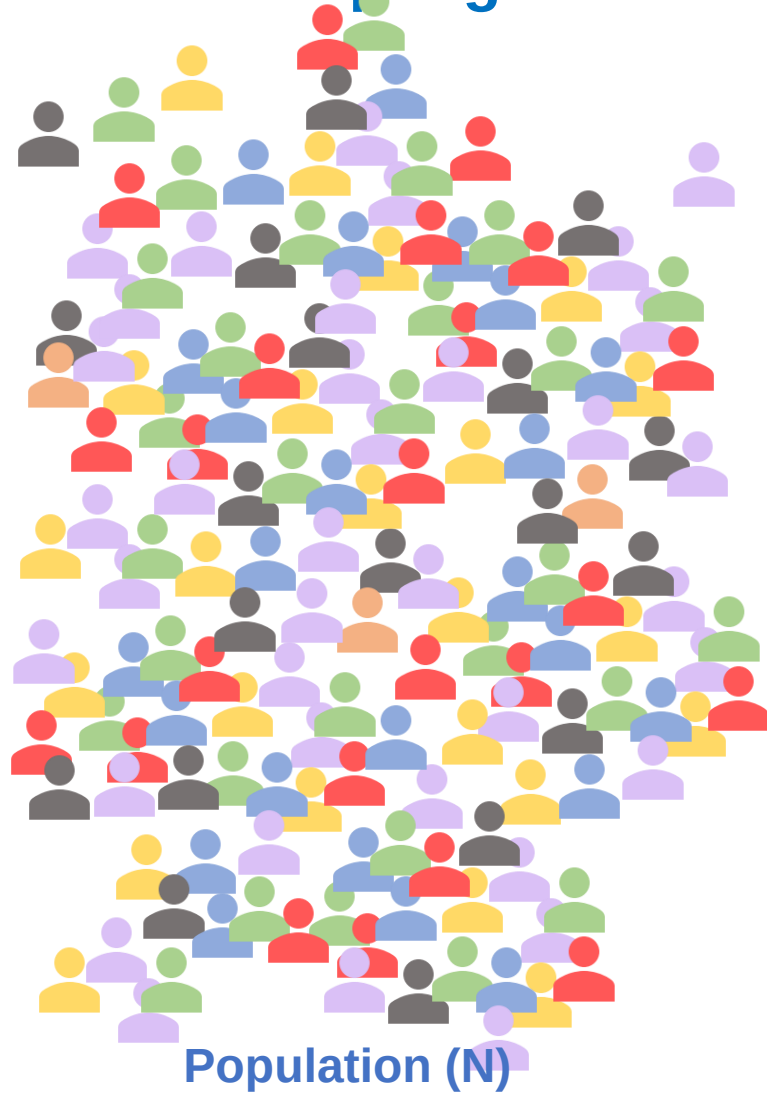
→ Group of people you are interested in.

Sample is the specific group of individuals **chosen from the target population** to participate in the study.

→ The smaller group we actually collect data from.

→ Selected based on certain criteria or sampling methods to ensure that it **accurately represents the larger population**.

What has sampling to do with statistics?



Sampling



Sample (n)

Descriptive Statistics

Inferential Statistics

What is the difference?

Descriptive statistics

Goal: Describing the data already available (sample data).

Example:

“What is the average age of our survey participants?”

“Does the average age of our participants differ by gender?”

Method:

- Absolute and relative frequencies
- Central tendency (Mean, Mode, Median), Shape
- Variability (Variance, Standard Deviation)



Inferential Statistics

Goal: Make inferences (draw conclusions) from the sample and generalize them to the population.

Example:

“Do women and men differ in their average income?”

“Are males more likely to gain a high salary than females?”

Method:

- Standard Error and Confidence Interval
- Hypothesis testing (Chi²-Test, t-test, ANOVA, Correlation)
- Linear- or logistic regression



What do I need to consider?

Descriptive statistics

Tell you **what your data look like**.

☞ “What is happening in my data?”

→ Works well with small samples and non-random sampling.

Inferential statistics

Tell you **whether your findings are likely a real phenomenon in your population or just coincidence**.

☞ “Is it statistically significant?”

→ Only works well with **sufficient sample size**

→ Small samples mean more uncertainty and less power to detect effects.

→ Requires **random sampling** to ensure the sample is representative of the population.

How do I know that my sample size is sufficient?



#...i think this meme demonstrated the importance of sample size better and more efficiently than any science class i've ever taken

Tools and Guidelines

[Australian Bureau of Statistics Sample Size Calculator](#)

[UK Sample Size Calculator](#)

[Gesis Sample Size Calculation For Complex Sampling Designs](#)

Programmes and packages

[G*Power](#) Desktop Application

R [pwr](#) and [practools](#) packages



What is a variable?

In quantitative analysis, we investigate **attributes** (like characteristics) of our observations (one person, case), e.g. age, gender, income, self-efficacy score, opinions

- Variables are the attributes we want to investigate
- Variables can vary between people and/or over time

Example:

What does the distribution of **age** look like in my sample?

What does the **age** distribution of the population in Germany look like on 2nd December 25?

Are **women** more likely to reach the **age** of 100 in Germany than men?

Does the mean **age** differ between **gender** groups in my sample?

Which **characteristics** influence the **age** people reach in Germany?

Definition

How is each variable **measured**?

What is the **mathematical nature of the values** assigned to each variable?

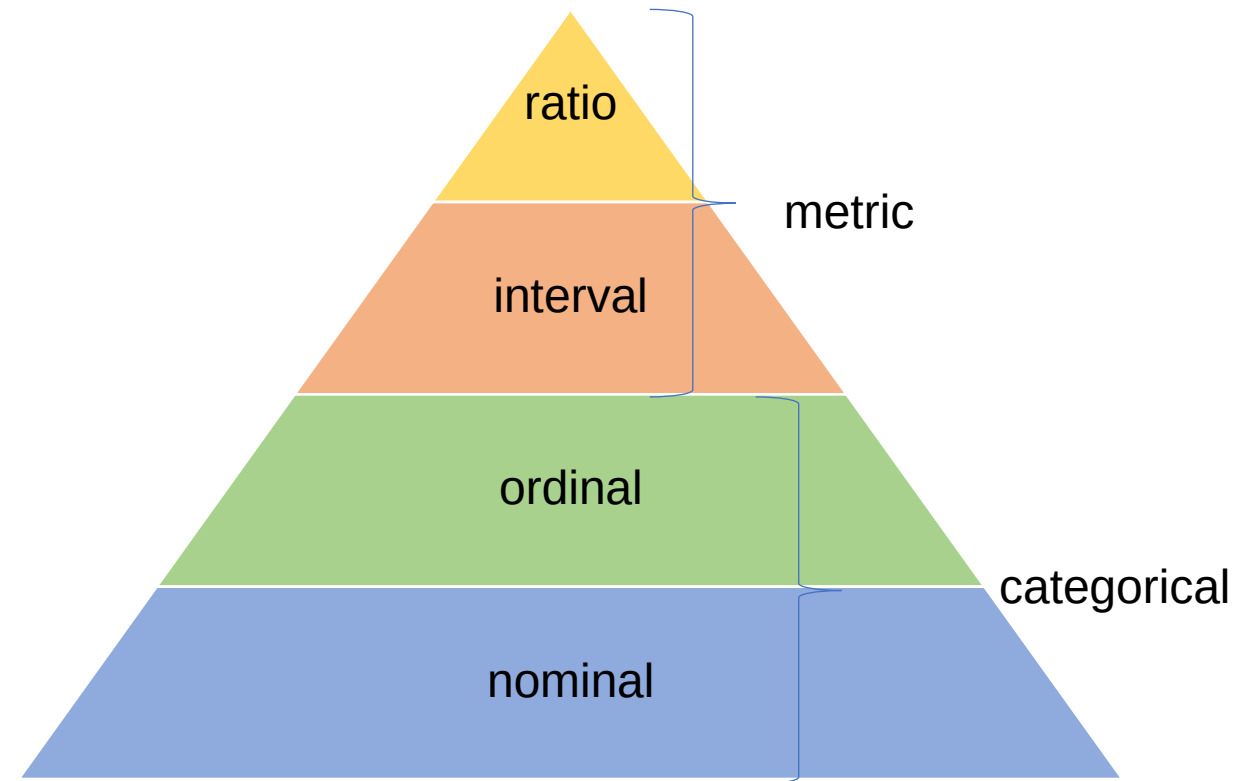
Importance

Level of measurement determines the type of **statistical analysis** you can carry out on the outcomes of your questions.

These levels reflect how data can be:

- Quantified
- Analyzed
- Interpreted

Levels of Measurement



Examples

Nominal Data: mutually exclusive, labelled categories.

→ e.g. eye color, citizenship, marital status, mode of transportation



Ordinal Data: categories have a natural order or rank.

→ e.g. education level, school grade



Interval Data: Data is measured along a numerical scale that has equal interval between adjacent values

→ e.g. temperature in Celsius, IQ score

Ratio Data: Data is measured along a numerical scale that has equal interval between adjacent values and a true zero.

→ e.g. weight in KG, number of staff, income in Euro

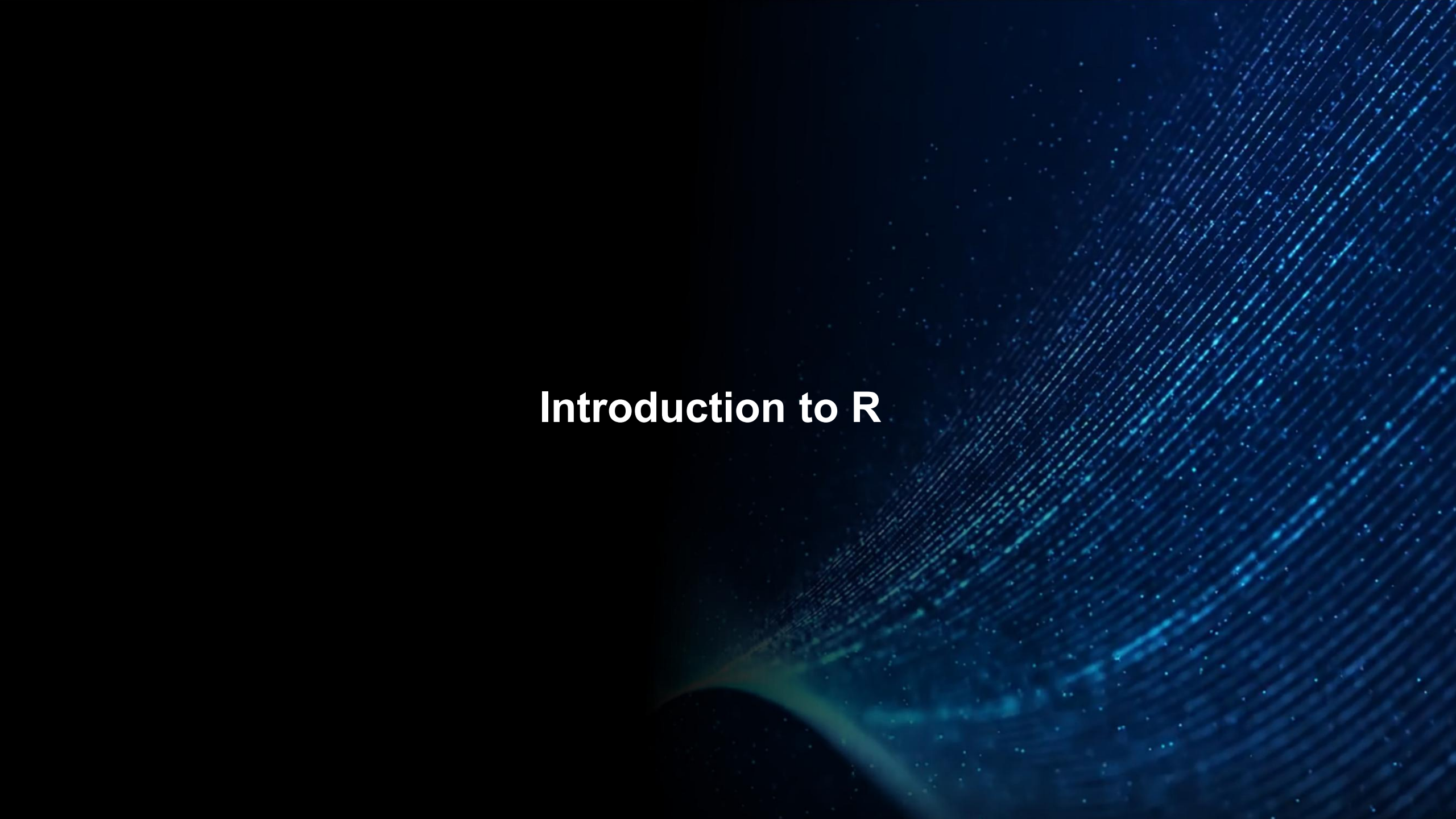


Comparison

THE FOUR LEVELS OF MEASUREMENT:

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

Introduction to R



Your turn!

1. Copy folder `Materials` from to your local machine.
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 - Path to subfolder `Exercises` will be your **working directory**.
 - Folder content: data sets, exercises without & with solution
2. Do exercise **`00_Introduction_to_R`**.



<https://de.pngtree.com>

Data set structure

Observation/Case/Record:

One row in the dataset, usually one respondent or unit.

Variable:

One measured characteristic/attribute

Value:

The specific entry for a variable in one case (e.g., 1)

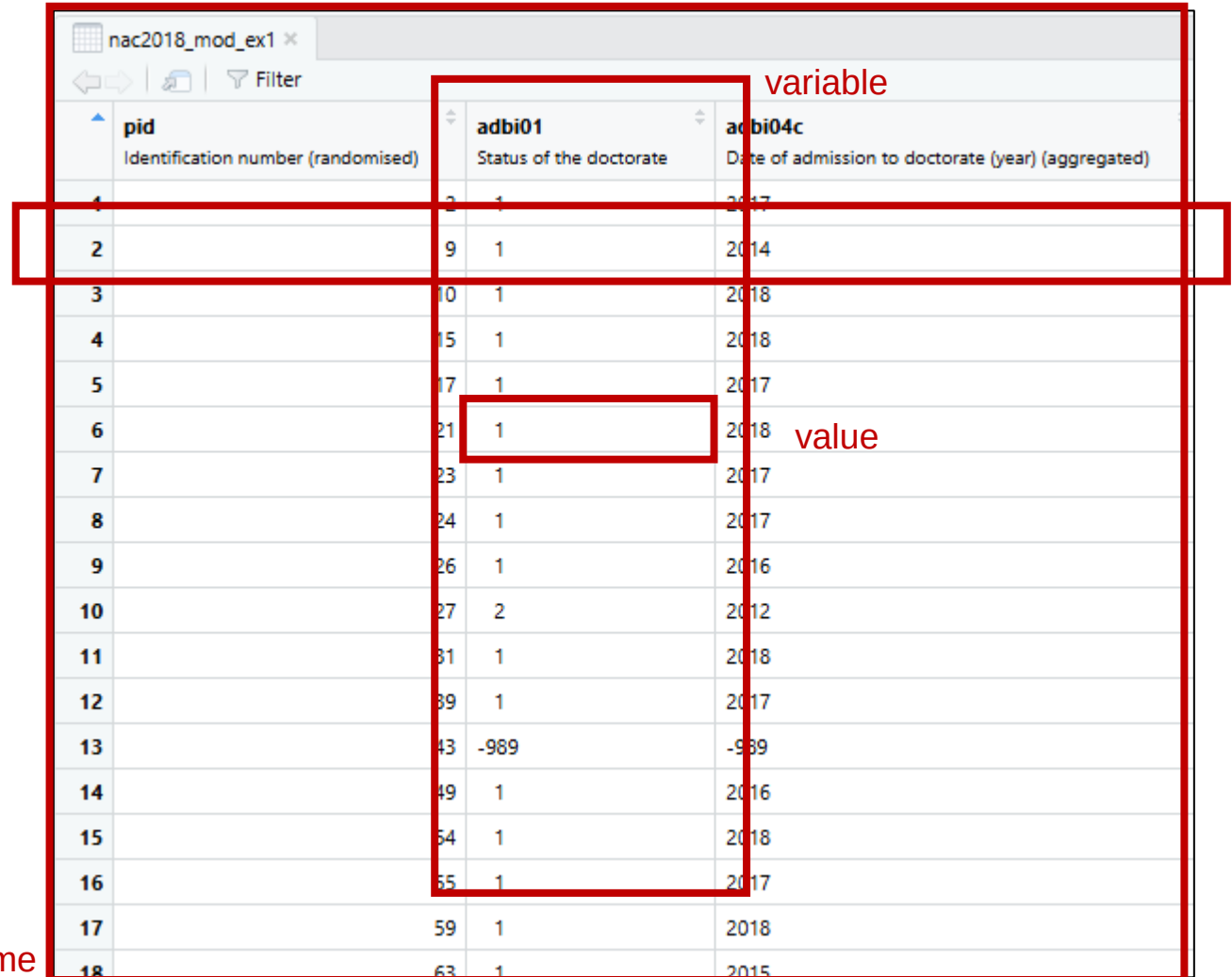
Data frame:

Dataset, the full table:

- rows = cases
- columns = variables
- cells = values

Observation/case/
record

Data frame



pid	adbi01	adbi04c
Identification number (randomised)	Status of the doctorate	Date of admission to doctorate (year) (aggregated)
1	2	1
2	9	1
3	10	1
4	15	1
5	17	1
6	21	1
7	23	1
8	24	1
9	26	1
10	27	2
11	31	1
12	39	1
13	43	-989
14	49	1
15	54	1
16	55	1
17	59	1
18	63	1

Example: Data frame in data viewer using `View()` function

Elements of a variable

Variable name:

The short technical name used in the dataset (e.g. adem01).

Variable label:

A descriptive text for the variable (e.g. "Gender").

Value:

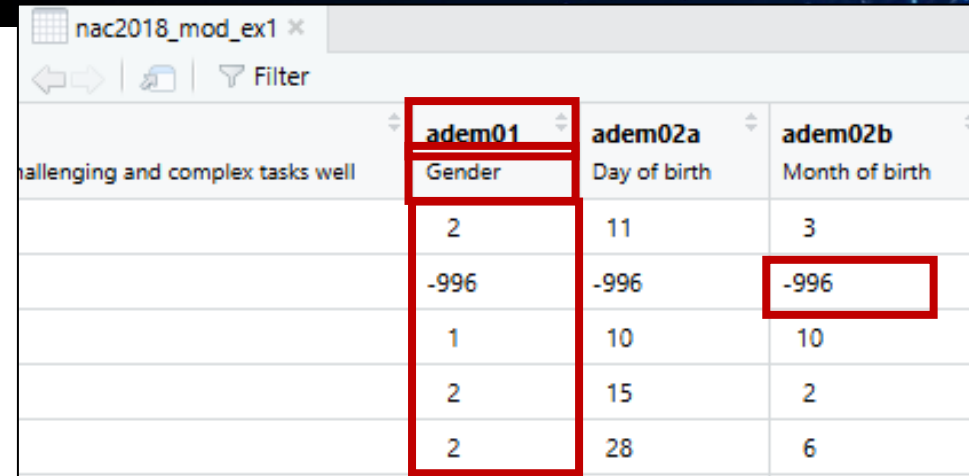
The specific entry for a variable in one case (e.g., 1).

Value label:

A description of what a value means (e.g., 1 = female).

Missing value:

Data not available, either left empty or coded (e.g. -996 = interview break-off).



challenging and complex tasks well	adem01	adem02a	adem02b
	Gender	Day of birth	Month of birth
	2	11	3
	-996	-996	-996
	1	10	10
	2	15	2
	2	28	6

```
> frq(nac2018_mod_ex1$adem01)
```

Gender (x) <numeric>

total N=7089 valid N=7089 mean=-157.19 sd=364.49

value	Label	N	Raw %	Valid %	Cum. %
-998	no answer	0	0.00	0.00	0.00
-996	interview break-off	1026	14.47	14.47	14.47
-989	filtered	43	0.61	0.61	15.08
-969	unknown missing	0	0.00	0.00	15.08
-966	not determinable	61	0.86	0.86	15.94
1	female	2916	41.13	41.13	57.07
2	male	3034	42.80	42.80	99.87
3	other	9	0.13	0.13	100.00
<NA>	<NA>	0	0.00	<NA>	<NA>

Example: Frequency table of variable `adem01` measuring gender using `frq()` function

Levels of Measurement & types and classes in R

Types are the basic **storage mode** of an object in R.

The class tells R **how to treat an object** and which methods apply.

Level of measurement	Type in R	Class in R
Nominal	Double, integer	Factors
Ordinal		Ordered factors
Interval		Numeric, integer
Ratio		
	Character (strings), logical TRUE/FALSE (boolean)	Date, data.frame, matrix, array, list

→ R often imports data as **character**, **double**, or **integer** – other variable classes need to be created explicitly e.g. for nominal (**factors**), ordinal (**ordered factors**) or date/time (**Date**) variables.

→ **Characters** usually need to be **converted** into another data type so that R can handle the variables correctly in analyses.

Is there a life beyond the PhD?

In this workshop, we will work with one coherent example throughout all four parts.

We focus on **two outcomes of interest**:

- **Overall life satisfaction (0-10 scale)**
- **Children (yes/no)**

We will examine how these outcomes are related to different aspects of doctoral researchers' lives and backgrounds:

PhD and Work conditions

- Status of the doctorate
- Discipline
- Emotional support during PhD
- Perceived scientific pressure
- Gross income

Attitudes and Well-Being

- Satisfaction with work-life balance
- Relationship with parents
- Self-Efficacy

Demographics

- Gender
- Age in years
- Highest vocational degree of parents
- Country of birth
- Relationship status

National Academics Panel Study (Nacaps)

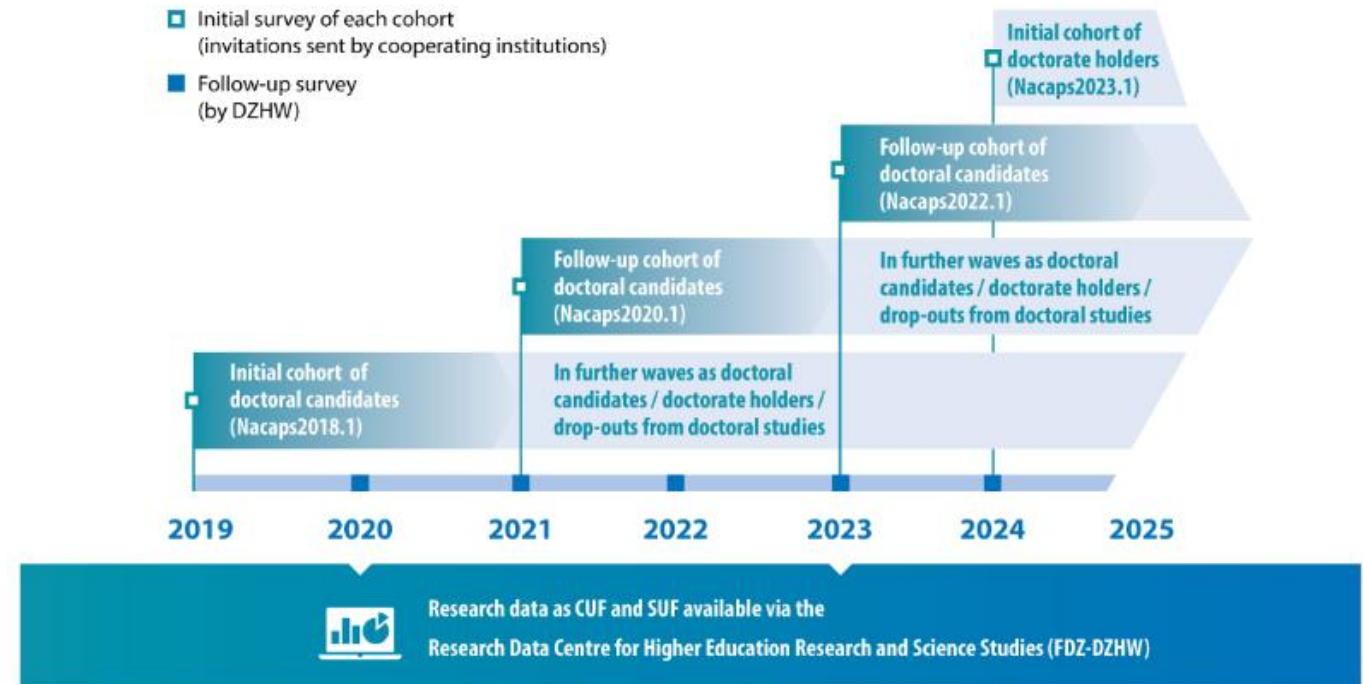
Project duration: Januar 2017 – ongoing (funded by BMBF)

Target population: full census survey

- All persons who started/completed a doctorate at a German HEI
- 66 participating universities

Survey design: multicohort longitudinal study

- Initial wave about one year after starting/graduating PhD
- Annual follow-up surveys
- New cohort every other year (doctoral candidates since 2018; doctorate holders since 2024)
- Online



Your research strategy

To explore these questions, we will:

1. start by **exploring the data** and getting to know our sample
2. look at **univariate** distributions of key variables (e.g. life satisfaction, number of children, income),
3. examine **bivariate** relationships (e.g. does life satisfaction differ between people of different phd status and gender)
4. finally fit **multivariate models**:
 - a **linear regression** with life satisfaction
 - a **logistic regression** on probability of having children

Your turn!

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 - Folder content: data sets, exercises without & with solution
 2. Do exercise “**01_qa_exploring_ex**”.
- Version with solution: **01_qa_exploring_ex_solution**



Univariate Statistics

The background of the slide is a dark, abstract composition. On the right side, there are numerous thin, bright blue lines that appear to be composed of small dots or particles, radiating outwards from a central point. At the bottom center, there is a soft, glowing greenish-yellow light that fades into the dark background.

What is it?

Univariate statistics deal with the **analysis of a single variable** at a time.
The goal is to describe and summarize its **distribution**.

Why univariate analyses are important:

- **Describe your sample**, that means summarize who or what you studied, e.g., age, gender, education.
- **Check data quality**, e.g. detect missing values, outliers.
- Check the **shape of the distribution**, e.g. skewness and kurtosis as a foundation for further analyses;
→ Many parametric tests, e.g. t-test or ANOVA, assume normally distributed variables.
- Provide **clear and interpretable** tables and charts for publications.

What decisions to make?

1. 🎯 Which variables to analyze?

Depends on your **research question**, for example:

- Your outcome(s) of interest,
- Attributes that might affect outcome(s) of interest, e.g. group differences, influencing factor,
- Demographics.

2. ⚙️ Which type of analysis and visualization to use?

Depends on the variable's **level of measurement**.

Level of measurement	Descriptive Analysis	Visualization
min. nominal	Absolute/relative frequencies, mode	Bar chart
min. ordinal	cumulative frequencies, median, percentiles	Bar chart
Interval/ratio	Mean, confidence intervals Variability (Standard, Deviation) Shape (Skewness, Kurtosis) Standard error and confidence interval	Boxplot, Histogram

Absolute and relative frequencies

Absolute frequencies (n): The number of cases in each category or value.

→ *How many?*

Relative frequencies (%): The proportion or percentage of cases in each category.

→ *How much out of the total?*

Example:

Gender	Absolute frequency (n)	Relative frequency (%)
Female	250	50
Male	200	40
Diverse/Other	50	10
Total	500	100

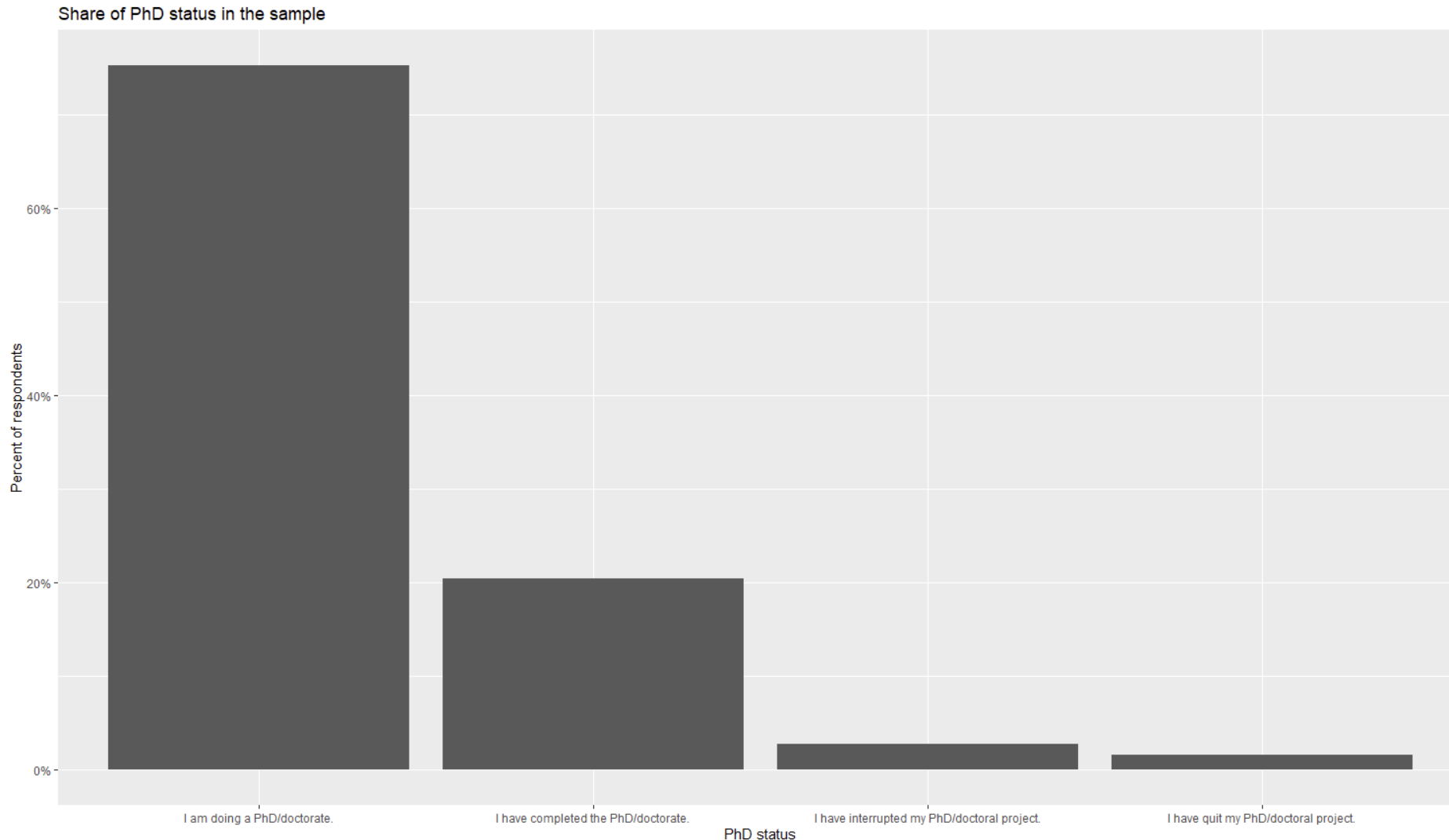
Frequency table reporting

Summarize absolute and relative frequencies in frequency table:

- Absolute (n) and relative frequencies (%) in columns;
- Variable categories in row;
- Cells contain absolute numbers and percentages in each category;
- Total row at the bottom to show the total sample corresponding to 100%.
 - Categories should add up to 100% 😊

Gender	n	%
Female	250	50
Male	200	40
Diverse/Other	50	10
Total	500	100

Visualization of absolute and relative frequencies: Bar chart



Example: Bar chart of variable `bdbi01` measuring Status of the doctorate using `ggplot()`

Measure of central tendency: Mode

The mode (in German: Modus) is the **most frequent value** of a variable.

→ Helps to describe the most typical or common category of a variable in your sample.

Example:

Gender	Absolute frequency (n)	Relative frequency (%)
Female	250	50
Male	200	40
Diverse/Other	50	10
Total	500	100

When to use:

Works well for **nominal** or **categorical** data.

Cumulative frequencies

Cumulative frequencies: Running total of absolute or relative frequencies.

→ *How many cases fall up to including a certain category?*

→ Useful only for variables where categories have a natural order.

Example:

Highest vocational degree	Absolute frequency (n)	Relative frequency (%)	Cumulative frequency (%)
no vocational qualification	25	5	5
Apprenticeship	250	50	55
Technical/master school	50	10	65
Bachelor/master degree	150	30	95
PhD	25	5	100
Total	500	100	

65 % of respondents finished technical/master school or lower, while 35 % have an university degree.

Percentiles

Percentiles divide an ordered distribution into 100 equal parts.

Each percentile shows the **value below which a certain percentage of cases fall**.

→ *How is the variable distributed?* — not just the average, but also *how spread out* it is.

→ They are based on cumulative frequencies, so they only make sense for ordinal or metric variables.

10th percentile (P10)

→ 10 % of cases are below or equal this value

25th percentile (P25)

→ 25 % of cases are below or equal to this value

50th percentile (P50 = **Median**) → half of the data below, half above

75th percentile (P75)

→ 75 % of cases fall below this value

} **quantils** (because they're $\frac{1}{4}$)

Minimum (min)

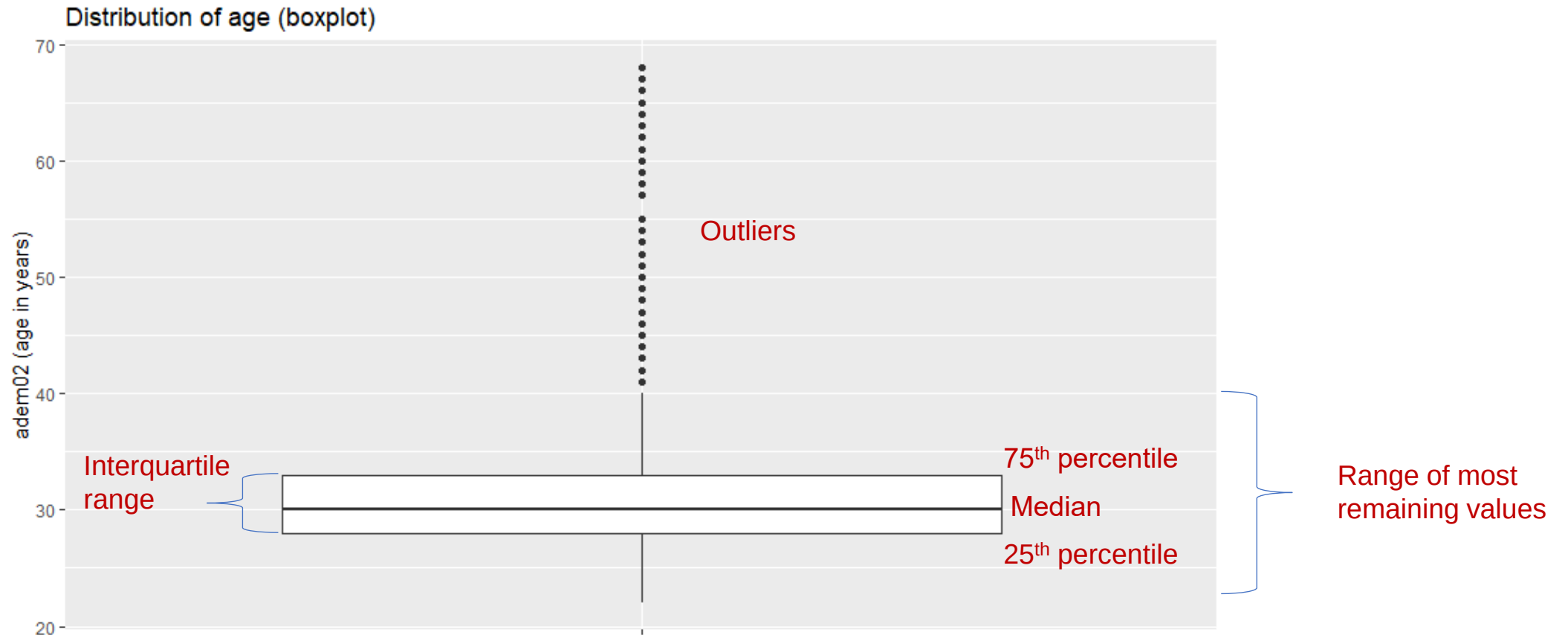
→ smallest observed value in a variable

Maximum (max)

→ largest observed value in a variable

} **range** of possible values

Visualization of percentiles: Boxplot



Example: Boxplot of variable `adem02` measuring age in years using `ggplot()`

Measure of central tendency: Mean ($\bar{x}$)

The Mean (in German: arithmetisches Mittel, Mittelwert) is the **arithmetic average** of all values:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of cases}}$$

Example:

Income (€)	Absolute frequency (n)
2000	1
2200	1
2800	1
3000	1
10000	1
Total (n)	5

$$\text{Mean} = \frac{2000 + 2200 + 2800 + 3000 + 10000}{5} = 4000$$

→ Careful: The mean is **sensitive to outliers**.

Measures of central tendency: When to use which measure?

Measure	Works for	Sensitive to outliers?	What it tells you
Mode	Nominal, ordinal, interval, ratio	✗ No	The most frequent value
Median	Ordinal, interval, ratio	⚠ Slightly	The middle value (50% below/above)
Mean	Interval, ratio	☑ Yes	The mathematical average

Variability: Standard Deviation (s or SD)

Typical deviation from the mean.

It shows, on average, **how far each value lies from the mean**, expressed in the **same unit** as the data (e.g. €)

Example

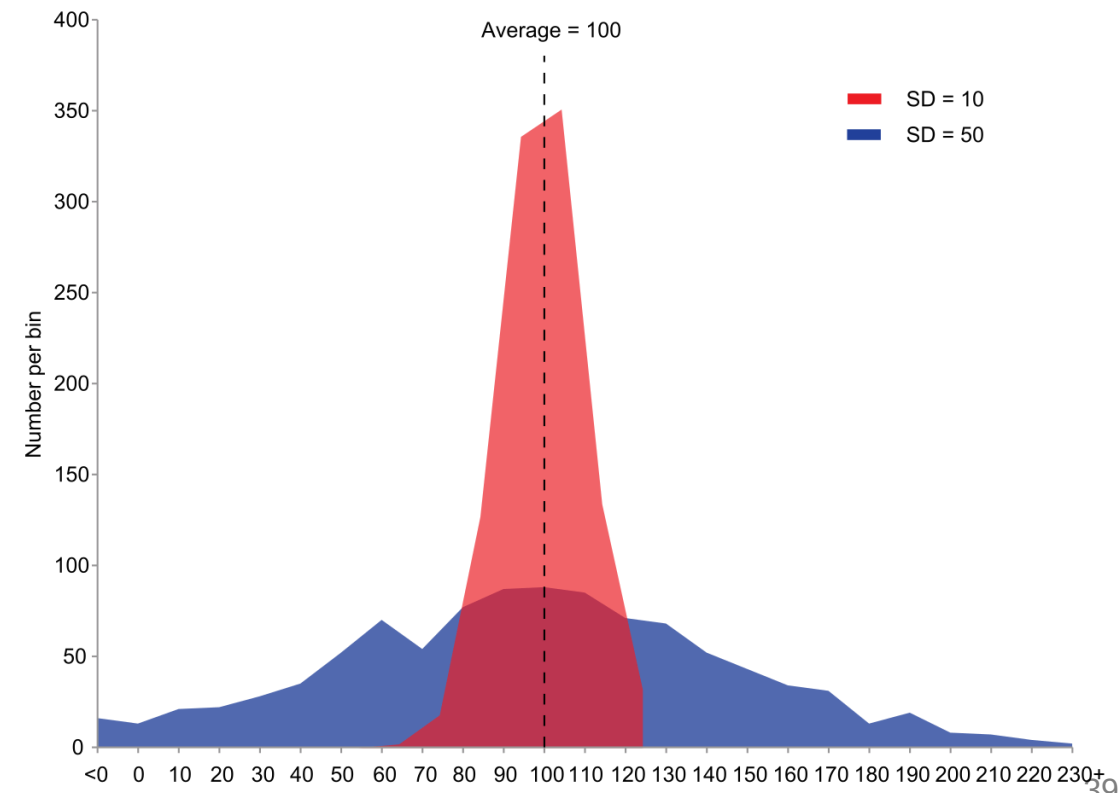
Sample of two population with the same mean but different standard deviations:

- red: small standard deviation
- blue: large standard deviation

→ Easy to interpret: “On average, values deviate from the mean by about s units.”

Small s → values are close to the mean
→ data are consistent / homogeneous

Large s → values are spread out widely
→ data are variable / heterogeneous



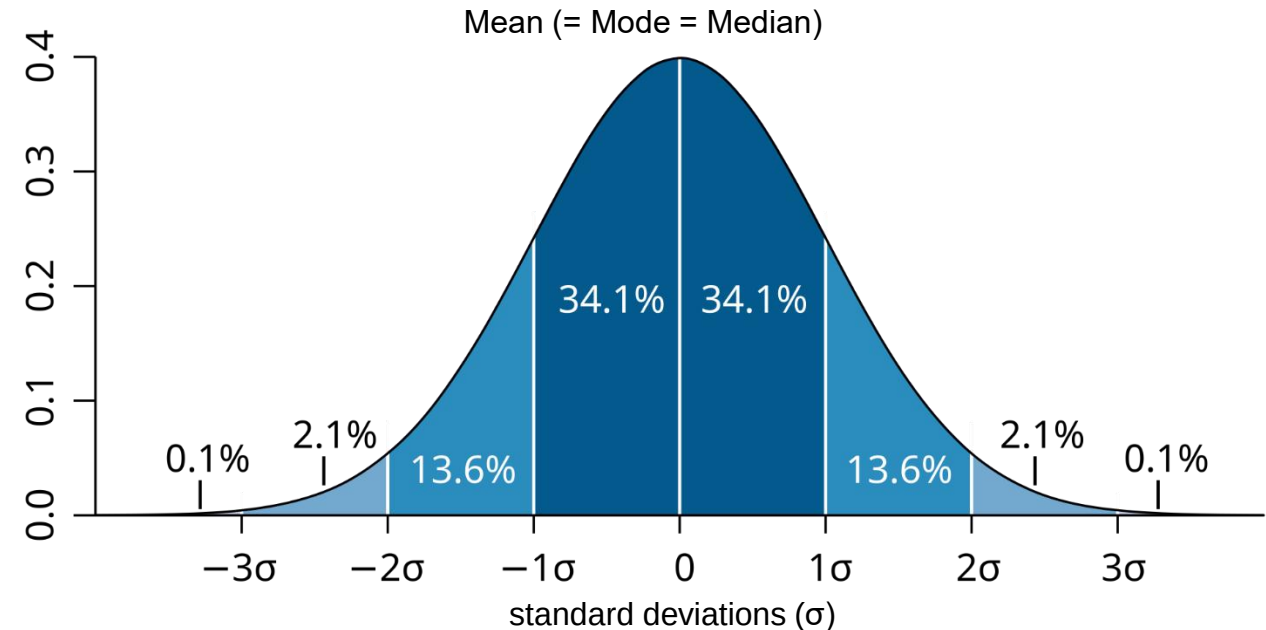
Normal distribution

The **normal distribution curve** (bell curve, *Normalverteilung*) represents how data are distributed when:

- most values are close to the **mean** and,
- fewer are very high or very low.

The peak in the middle marks the **mean**, which is also the **median** and **mode** in a perfect normal distribution.

- About 68% of all values lie within ± 1 standard deviation of the mean.
- About 95% lie within ± 2 standard deviations.
- About 99,7% cases fall within ± 3 standard deviations.



Why do I need to hear that?

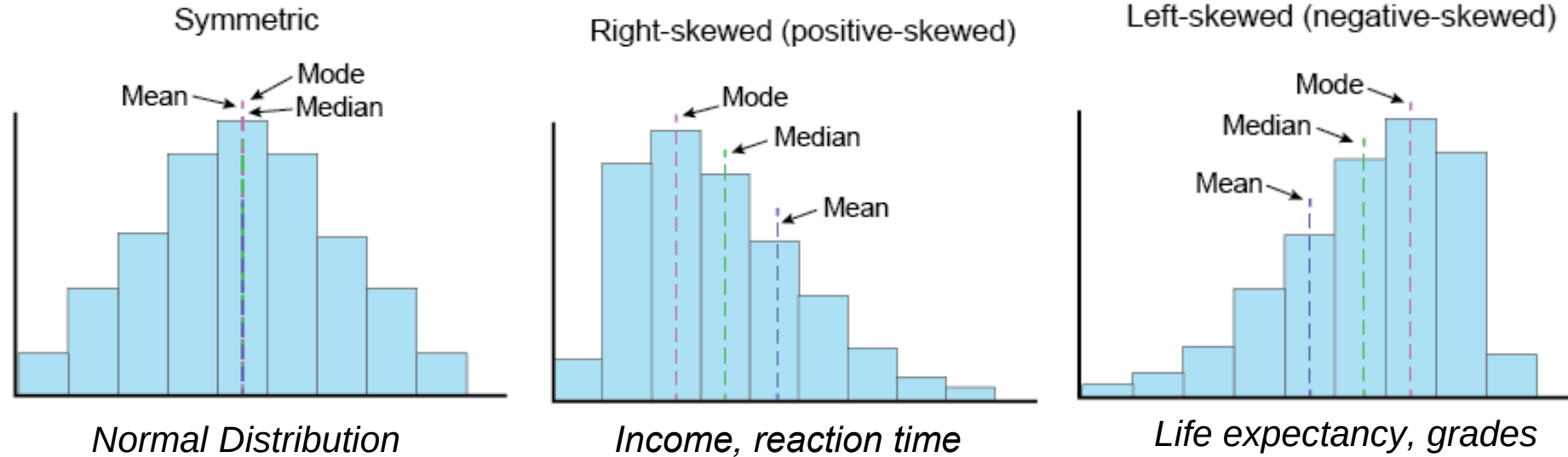
Many statistical inference tests (e.g. t-tests, ANOVA) assume normality



Skewness

Skewness (Schiefe) measures the **asymmetry** of a distribution:

- Symmetric
- Right-skewed
- Left-skewed



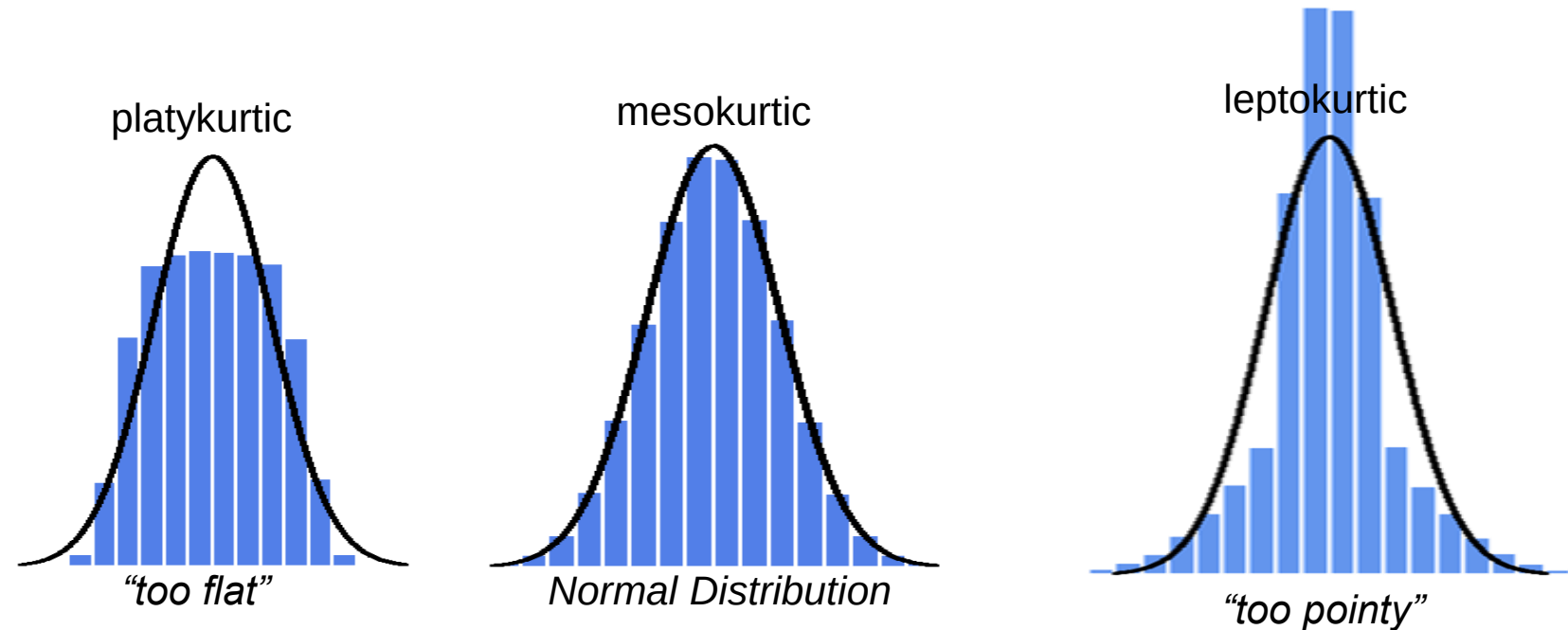
Interpretation

→ Tells you whether the values are more spread out on one side of the mean.

Skewness value	meaning
0	Symmetric, perfectly normal distribution
-0.5 – 0.5	Approximately symmetric
-1 – -0.5 and 0.5 – 1	Moderately skewed
-1 – 1	General acceptable for normal distribution
< -1 or > 1	Highly skewed

Kurtosis

Kurtosis (Wölbung) describes **peakedness** or **flatness** of a distribution compared to a normal distribution.



Interpretation

→ It describes how concentrated or spread out the values are around the mean.

kurtosis value	meaning
3.0	mesokurtic, perfectly normal distribution
< 3.0	platykurtic
> 3.0	leptokurtic

A kurtosis value between 2 to 4 (or -1 to +1 for excess kurtosis) is generally acceptable for normal distribution.

Variability: Standard Deviation and Variance

Standard deviation (s)

Typical deviation from the mean.

It shows, on average, **how far each value lies from the mean**, expressed in the **same unit** as the data (e.g. €).

→ But: negative deviations cancel each others out.

Example

values **2, 4, 6**, with a mean of **4**

Deviations from the mean:

- $(2 - 4) = -2$
- $(4 - 4) = 0$
- $(6 - 4) = +2$

If you simply sum them up: $-2 + 0 + 2 = 0$

Variance (s²)

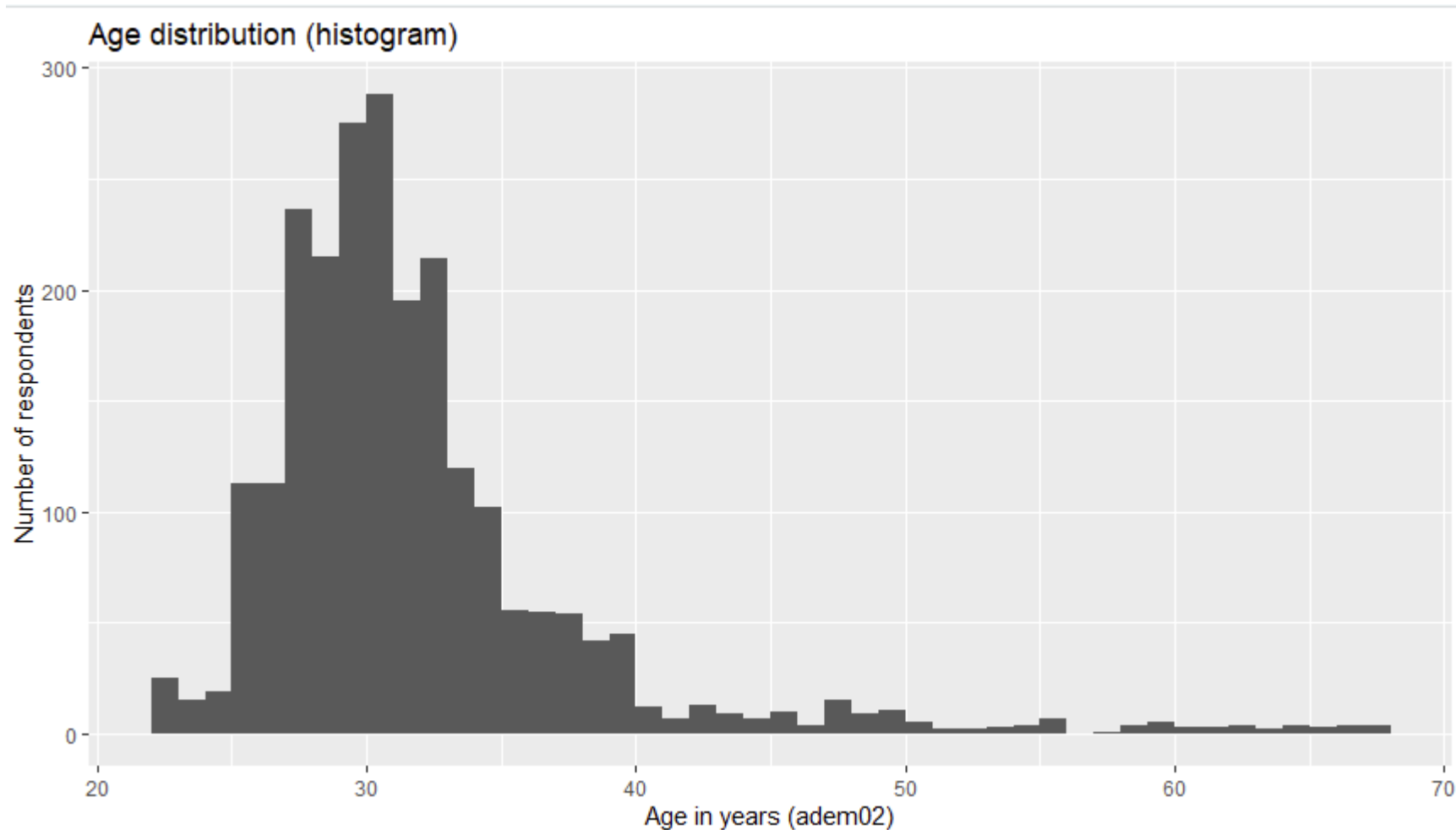
Average of squared deviations from the mean.

Measures how much the values differ from the mean **on average**, expressed as squared unit (e.g. €²).

→ Squaring prevents negative deviations from cancelling each other out.

→ Even if it's not easy to interpret directly (because of squared units), it gives us a **precise measure of overall variability** in a dataset.

Visualization: Histogram



Example: Histogram of variable `adem02` measuring age in years using `ggplot()`

Inference Statistics: Standard Error of the Mean ($SE_{\bar{x}}$)

How precise is our sample mean as an estimate of the population mean?

→ Smaller $SE_{\bar{x}}$ → more precise estimate

$$SE_{\bar{x}} = \frac{SD}{\sqrt{n}}$$

- If **SD is large** → SE is large (more variability → less precision).
- If **n is large** → SE is small (more data → more precision).

Standard deviation (SD, s)

Standard deviation describes how much **individual values** **vary** around the mean.

→ Spread of the data.



Standard Error (SE)

If we **repeated the study many times**, the sample means would vary. The standard error describes this variation.

→ Uncertainty of the estimate.

Inference Statistics: Confidence Interval (CI)

Instead of a single estimate, what is a *plausible range of values for the population mean*?

→ Narrow *CI* → more precise estimate

For a 95% CI of the mean:

$$\bar{x} \pm 1.96 * SE_{\bar{x}}$$

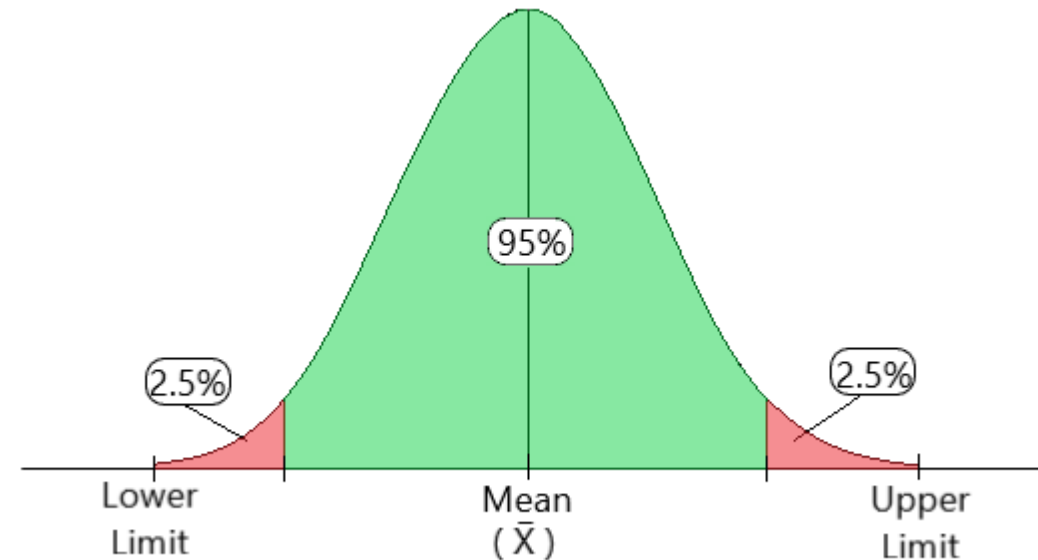
(or using a t-value for smaller samples from the t-distribution table)

Larger **SE** → **wider** CI.

Larger **n** → smaller SE → **narrower** CI.

There is no absolute rule for when a CI is 'narrow' or 'wide'.

It always has to be judged relative to the measurement scale and the research question.



How to report Mean, SD and CI?

 **Report always both, the mean and SD (standard deviation).**

- Standard deviation is preferred over variance because it's easier to interpret
- Variance (s^2) is usually only reported when it's directly needed for another calculation or method (e.g., ANOVA).

 Whenever the mean is an important result, **report it together with the SD and a 95% confidence interval** to show how precise the estimate is.

- Standard error (SE) is mainly a technical quantity used to construct the CI and is rarely reported on its own.

Example: $M = 13.2$ ($SD = 0.8$, 95% CI [13.04, 13.36], $n = 100$)

→ The average wage/hour is 13,30 €. On average, values differ from the mean by 0,80 €.

→ A 95% confidence interval for the population mean ranges from about 13.0 € to 13.4- €.

Your turn!

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 - Folder content: data sets, exercises without & with solution
2. Do exercise “**02_qa_uni_ex**”
Version with solution: **02_qa_uni_ex_solution**



Bivariate Statistics

The background of the slide is a dark blue gradient. On the right side, there are numerous thin, bright blue lines and dots that create a sense of motion, resembling a particle trail or a data visualization. These lines curve upwards and to the right, adding a dynamic feel to the otherwise static text.

What is it?

Bivariate statistics deal with the analysis of the **relationship between two variables at a time**.

Why bivariate analyses are important:

- **Explore relationships**, how one variable changes with another (e.g., age and income, education and job satisfaction).
- **Compare groups**, e.g. mean differences between gender groups, income groups.
- **Check assumptions** before running multivariate models (e.g., linearity, homoscedasticity, independence).
 - Many inferential methods (e.g., regression, ANOVA, chi-square tests) **start with bivariate exploration**.
- Provide **clear and interpretable** tables and charts for publications.

What decisions to make?

1. 🎯 Which variables to analyze together?

Depends on your **research question**, for example:

- **Are there differences between groups?** e.g., gender → mean income
- **Do two variables move together?** e.g., correlation age and interest in politics
- **Is there an association between categories?** e.g., education level × employment status

2. ⚙️ Which type of analysis to use?

Depends mainly on the **level of measurement**

Variable X + Variable Y	Suitable analysis	Visualization
Categorical + categorical	Cross tabulations, Chi-square test	Clustered bar chart
Categorical + metric	Mean group comparison, t-test, ANOVA	Grouped box plot
Metric + metric	Correlation, linear regression	Scatterplot

Cross tabulation (Crosstab)

A **cross tabulation** (or contingency table, Kreuztabelle/Kontingenztabelle) summarizes the relationship between two categorical variables by showing **how often each combination of categories occurs**.

- Identifying patterns and associations between two categorical variables e.g., *Is education level related to employment status?*
- Comparing group differences → e.g., *Does voting preference vary by gender?*

Gender	Left		Greens		SPD		CDU/CSU		Other		Total
Women	70 (10.8% / 50.7%) +	+	200 (30.8% / 59.7%)	+	180 (27.7% / 54.5%)	+	150 (23.1% / 35.5%)	+	50 (7.7% / 40.0%)	=	650 100%
Men	60 (9.2% / 43.5%) +		120 (18.5% / 35.8%)		140 (21.5% / 42.4%)		260 (40.0% / 61.6%)		70 (10.8% / 56.0%)		650
Diverse	8 (16.0% / 5.8%)		15 (30.0% / 4.5%)		10 (20.0% / 3.0%)		12 (24.0% / 2.8%)		5 (10.0% / 4.0%)		50
Total	138 100%		335		330		422		125		1,350

Example: Voting behaviour and gender; absolute and relative frequencies absolute count (row % / column %).

Crosstab reporting

Ask yourself which variable is the **grouping variable**.

1. If you want to understand **effects** or **group differences**:

group = row → use row %

Example: gender differences in voting behaviour

Gender	Left	Greens	SPD	CDU/CSU	Other	Total
Women	70 10.8%	200 30.8%	180 27.7%	150 23.1%	50 7.7%	650
Men	60 9.2%	120 18.5%	140 21.5%	260 40.0%	70 10.8%	650
Diverse	8 16.0%	15 30.0%	10 20.0%	12 24.0%	5 10.0%	50
Total	138 100%	335 100%	330 100%	422 100%	125 100%	1,350

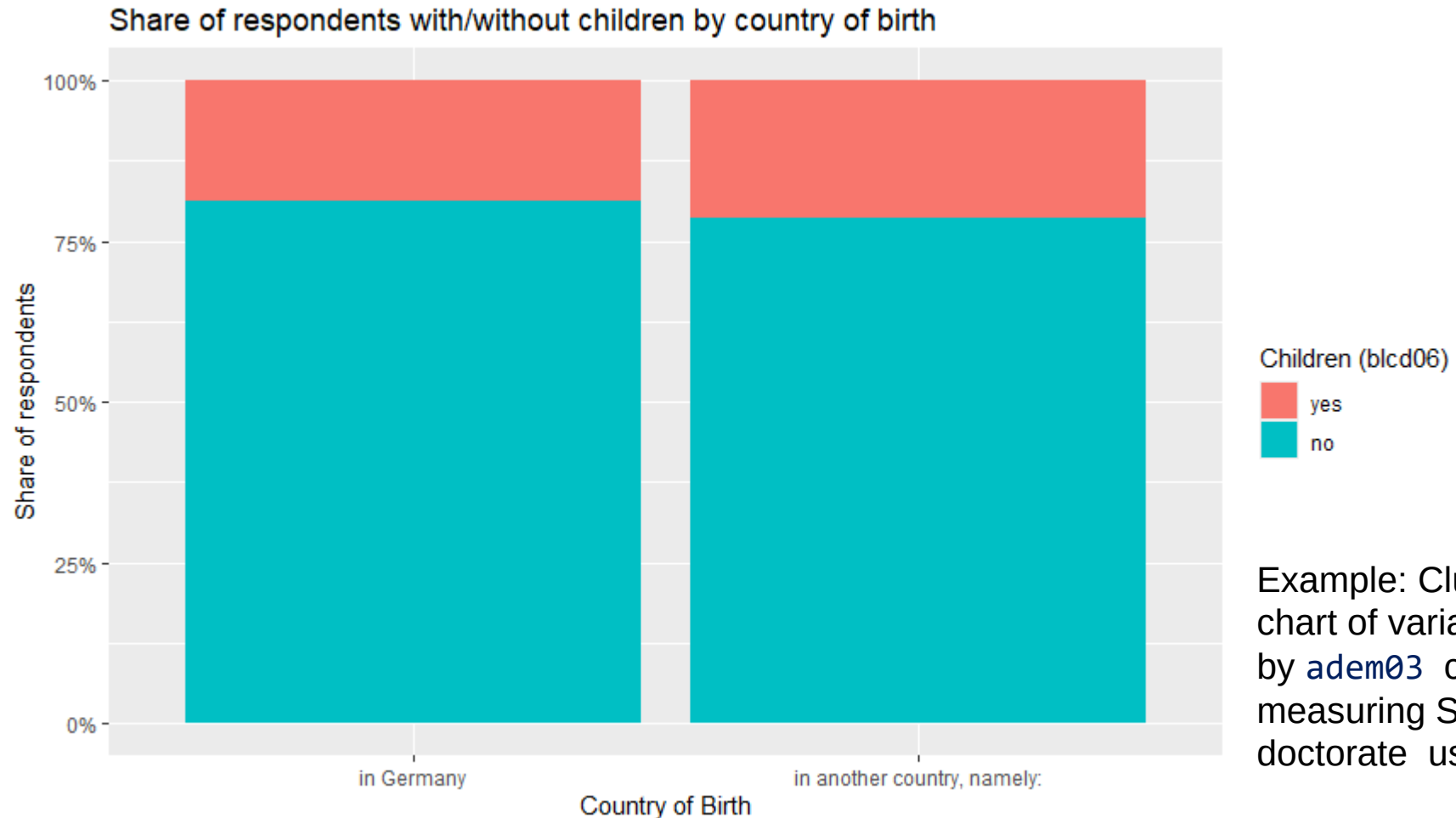
2. If you want to understand **composition**:

group = column → use column %

Example: gender composition of voters

Gender	Left	Greens	SPD	CDU/CSU	Other	Total
Women	70 50.7%	200 59.7%	180 54.5%	150 35.5%	50 40.0%	650 100%
Men	60 43.5%	120 35.8%	140 42.4%	260 61.6%	70 56.0%	650 100%
Diverse	8 5.8%	15 4.5%	10 3.0%	12 2.8%	5 4.0%	50 100%
Total	138	335	330	422	125	1,350

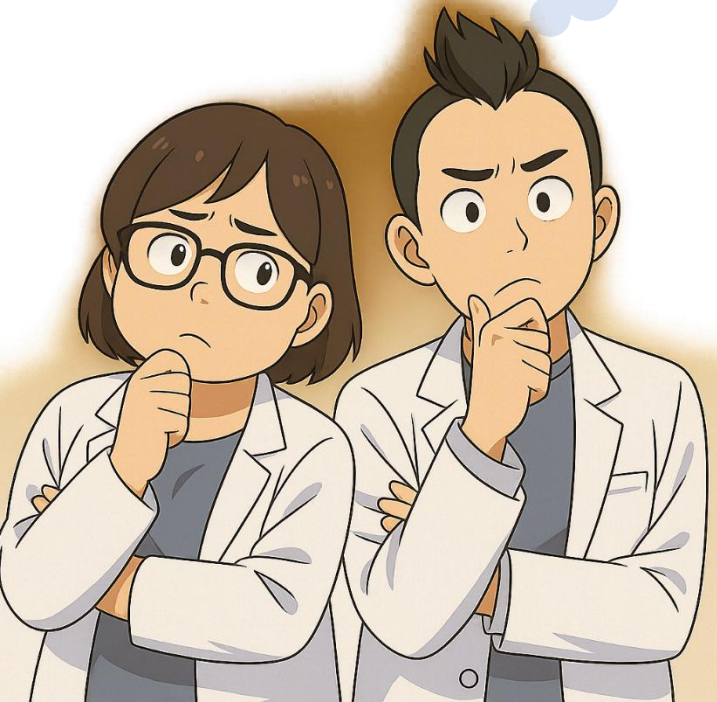
Crosstab Visualization: Clustered bar chart



Example: Clustered Bar chart of variable `blcd06` by `adem03` country of birth measuring Status of the doctorate using `ggplot()`

Inference Statistics: Hypothesis testing

How can I know that the pattern in my crosstab is real and not just coincidence?



Alternative hypothesis (H_1)

There is a relationship between the two variables.
(The differences are not random.)

Null hypothesis (H_0)

There is no relationship between the two variables.
(Any differences in the table are just random.)

Chi-square test (χ^2 -test) of independence

The Chi-square (χ^2) test tests for **statistical significance** in cross tabulations.

→ The Chi-square test tells us whether we should **reject H_0**

What does the Chi-square test do? (in very simple words)

- The table shows the **observed counts** (the real data).
- The test calculates what the counts **would look like if there were no relationship** between the variables (expected counts).
- Then it compares the two.

→ The **bigger** the difference between observed and expected χ^2 , the more likely it is that the **relationship is real**.

Chi-square test: p-value

The p-value tells you **how likely it is to see your result just by chance** if there is no real effect.

$p < 0.05$ means: There is **less than a 5% chance** that the pattern in your data is just random.

Small p-value (**$< .05$**) → very unlikely to be random

→ We reject H_0 , there is a **significant relationship**.

Large p-value (**$> .05$**) → easily could be random

→ **No significant relationship**

Degrees of freedom (df) show how many independent pieces of information are available for estimating variability.

→ They adjust the test so that p-values are accurate.

Example sentence for reporting:

The gender differences in voting behavior were significant, $\chi^2(8) = 21.5$, $p = .006$.

Mean group differences

Mean group differences analyze whether the **average value of a metric variable differs across groups of a categorical variable**.

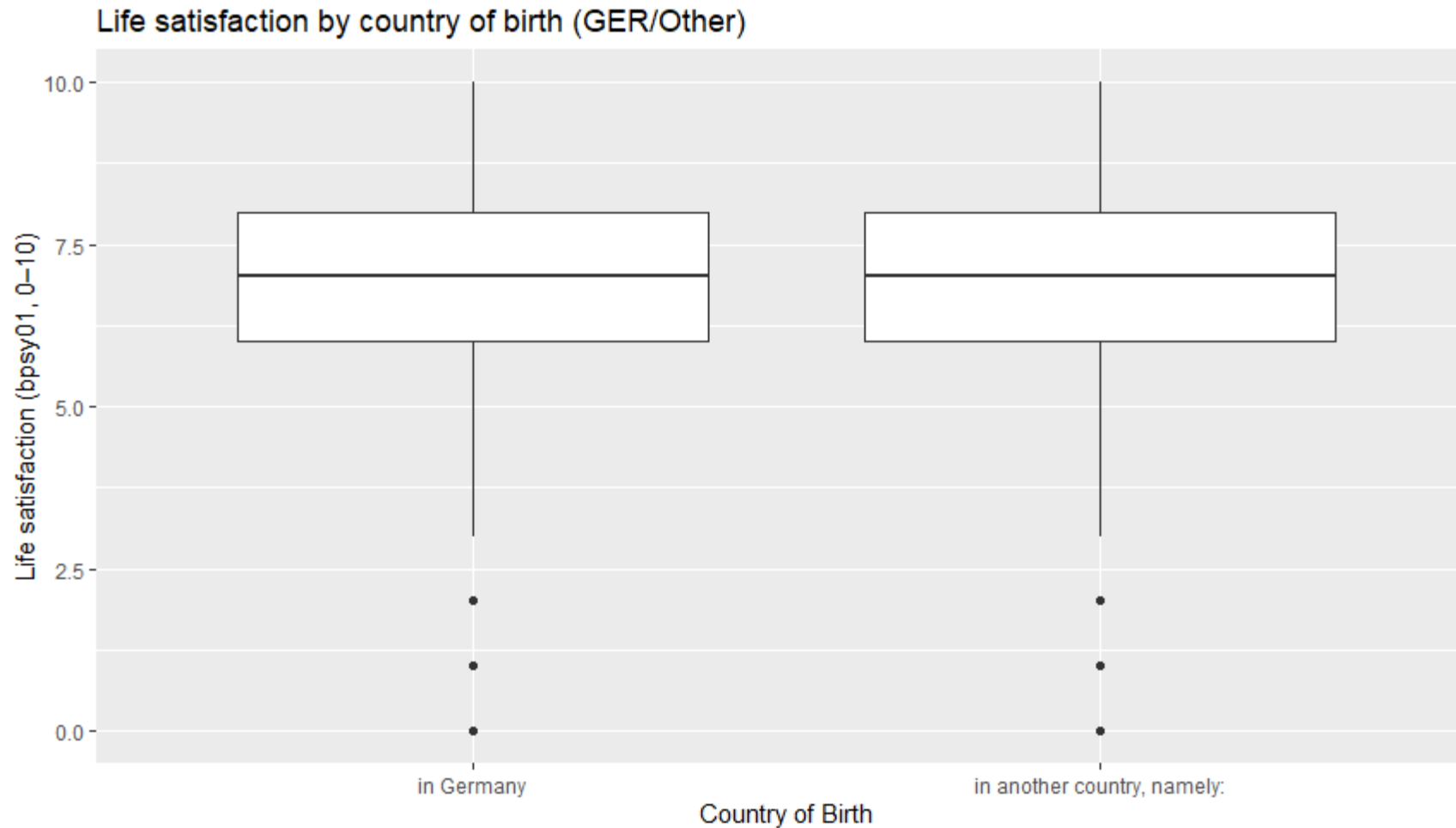
→ *Do education groups differ in their voter turnout?*

→ *Do gender groups differ in average income in €?*

Example: gender differences in monthly average gross income in €

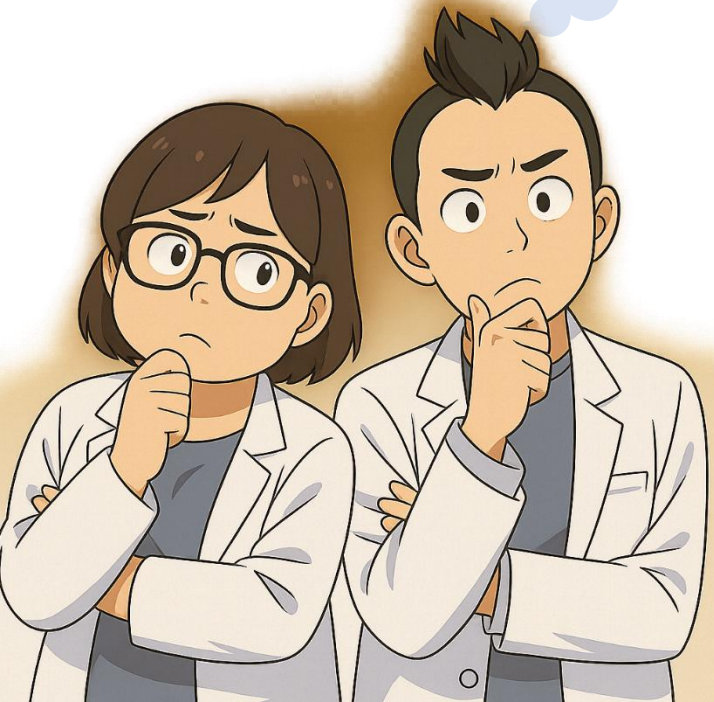
Gender	Mean income (€)	SD (€)	n
Women	2,900	650	650
Men	3,200	850	650
Diverse	3,050	700	50

Mean group differences visualization: Grouped box plots



Inference Statistics: Hypothesis testing

How can I know that the differences in means between groups and not just coincidence?



Alternative hypothesis (H_1)

There is a relationship between the two variables.
(The differences are not random.)

Null hypothesis (H_0)

There is no relationship between the two variables.
(Any differences between groups are just random).

T-test and ANOVA: Assumptions

T-test and ANOVA require:

- Approximately **normal distribution** within each group.
- Homoscedasticity: equal variances, that means groups have **similar spread**.
→ If not, use Welch t-test or Welch ANOVA (more robust).
- No strict **minimum sample size**, BUT small samples = unstable results, low power, and sensitivity to non-normality.
→ T-test and ANOVA work best with:
 - At least 20-30 cases per group (independent) or pairs (paired);
 - Similar group size.
- To detect small differences, you often need larger samples.

T-test and ANOVA

Testing for **statistical significance** in mean differences across groups.

T-test

Compares means of **two groups**

Are the two group means different beyond what we would expect by chance?

→ Looks at the **difference between the means** relative to the **variation within the groups**.

Independent t-test

Compares means of **two separate groups** that have **different participants**.

- Employed vs unemployed
- Treatment group vs. control group

Paired t-test

compares means **within the same individuals/paired units**.

- Before vs. after an intervention (same participants)
- Matched pairs (e.g., each patient matched with a control)

(oneway) ANOVA (F-test)

Compares means of **three or more groups**

Do at least one of these group means differ significantly?

- Male, female, diverse
- Scientific discipline

→ Looks at the **difference between the means** relative to the **variation within the groups**.

→ ANOVA only tells you **that a significant difference exists, not which groups differ**.

T-test and ANOVA: p-value

The p-value tells you how likely it is that the **difference we observe between group means** is just **due to random chance**.

→ Test become significant if the **between-group variance is much larger than the within-group noise**.

$p < 0.05$ means: There is **less than a 5% chance** that the differences in group means in your data is just random.

Example sentence for reporting:

The mean voting turnout differed significantly between employed and unemployed, $t(128) = 2.45$, $p = .016$.

t-test: $t(df) = \text{value}$, $p = \dots$

Example sentence for reporting:

The mean income differed significantly across gender groups, $F(2, 147) = 4.21$, $p = .017$.

ANOVA: $F(df1, df2) = \text{value}$, $p = \dots$

Pearson's correlation (r)

Correlation analyses examine whether two metric variables move together and if changes in one variable are systematically associated with changes in the other.

→ Do older people report higher political interest?

→ Do hours studies predict exam scores?

Scatterplot

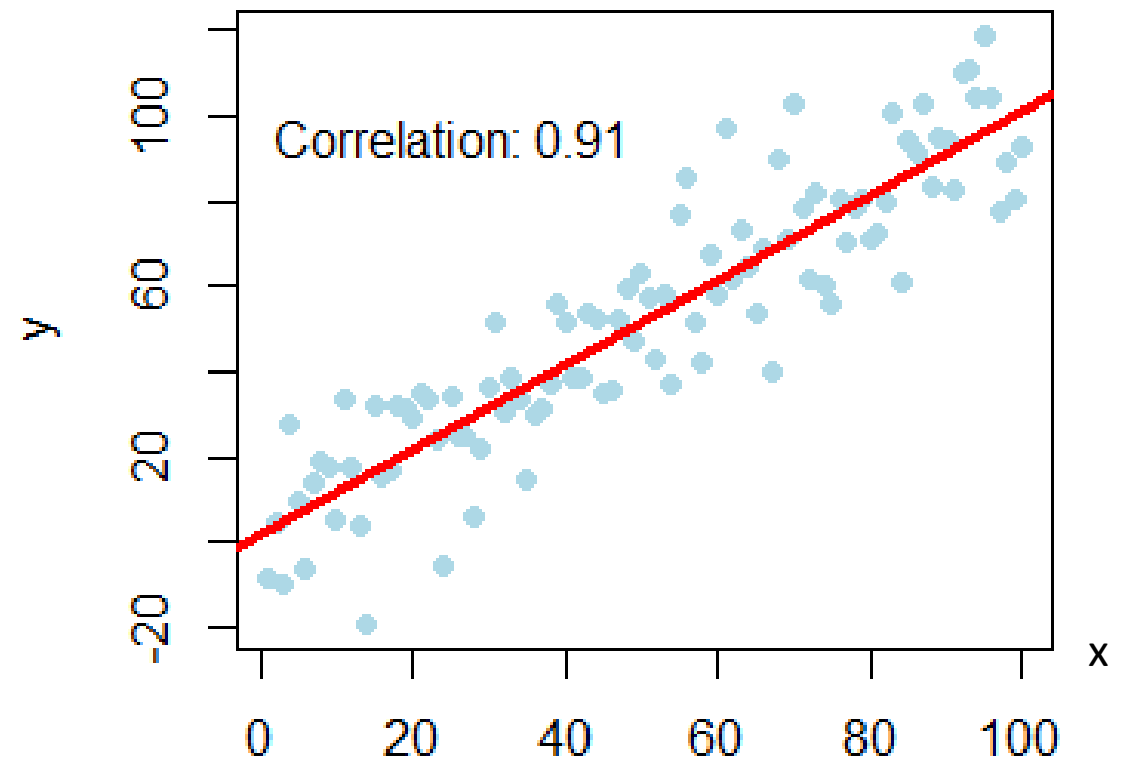
Each point represents one person/case;

X-axis = variable 1;

Y-axis = variable 2;

Shows the pattern: positive, negative, or zero;

Add a regression line to show the trend.



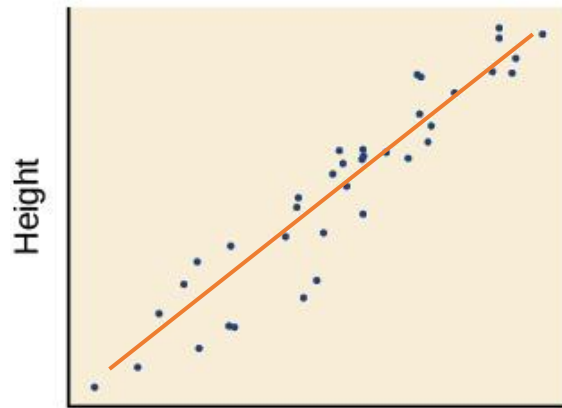
Pearson correlation (r): Assumptions

Pearson correlation require:

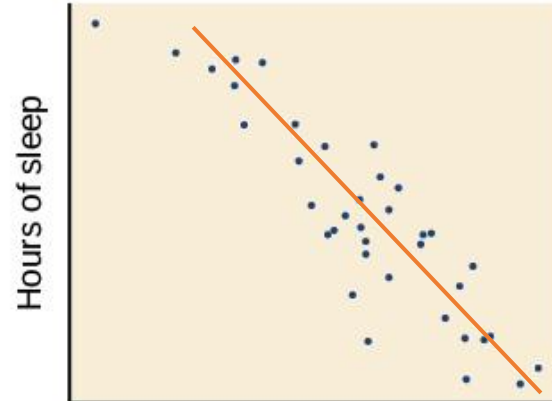
- Both variables are **metric**;
→ Use Spearman correlation for ordinal variables.
- **Linear** relationship;
→ The relationship should look roughly like a straight line in a scatterplot.
- **No strong outliers**;
- Approximately **normal distribution**;
- **Homoscedasticity**: the spread of Y should be fairly similar across values of X;
→ Equal variance” of points along the line.
- **Independent** observations;
→ Each case/person must represent an independent data point.

Pearson correlation (r) interpretation

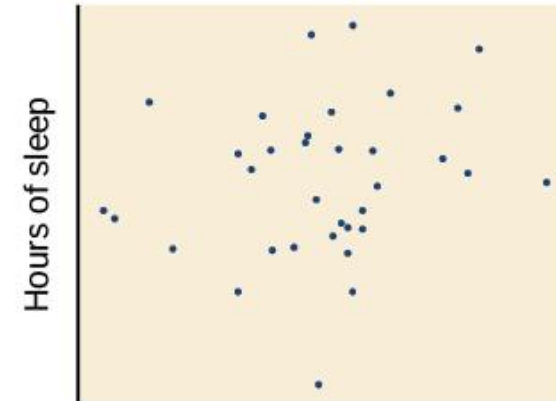
r tells you whether people who have more of X also tend to have more (or less) of Y.



Weight
(a) Positive Correlation



Tiredness
(b) Negative Correlation



Shoe size
(c) No Correlation

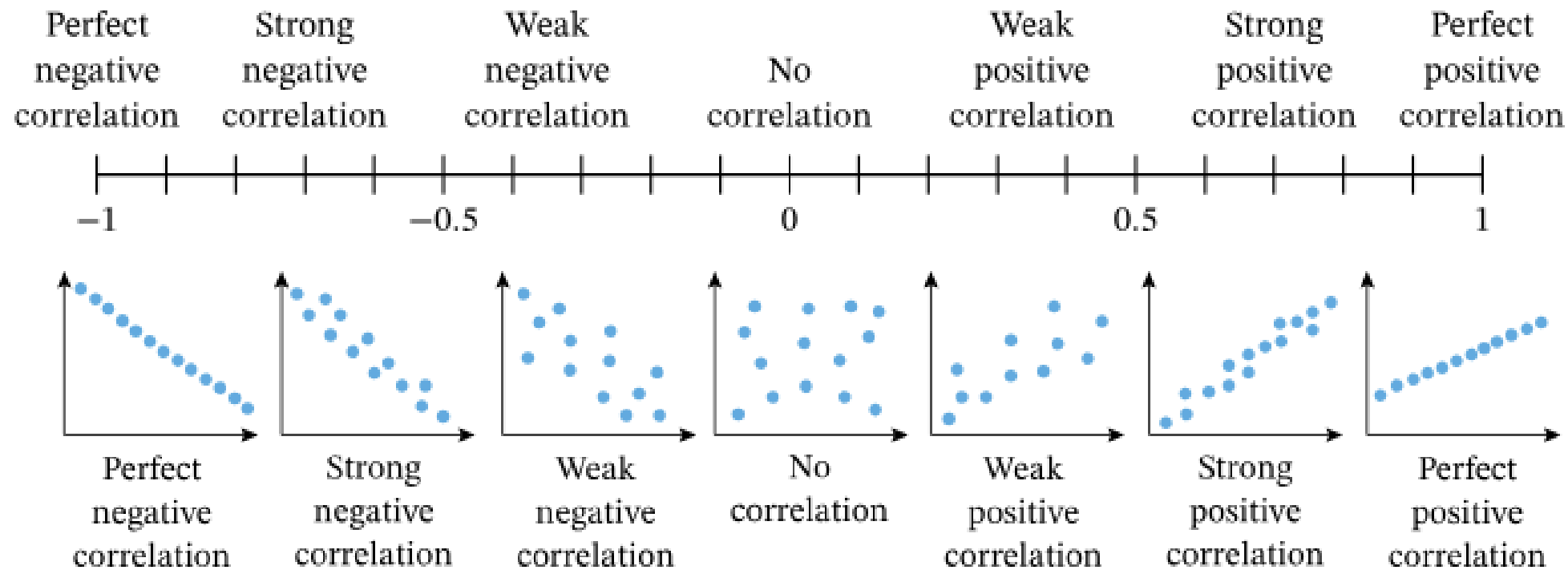
<https://opened.cuny.edu/courseware/lesson/16/student/?section=2>

The correlation coefficient r ranges from -1 to $+1$

- $r > 0$: when X increases, Y tends to increase (positive relationship).
- $r < 0$: when X increases, Y tends to decrease (negative relationship).
- $r \approx 0$: no systematic relationship.

Pearson correlation (r) interpretation

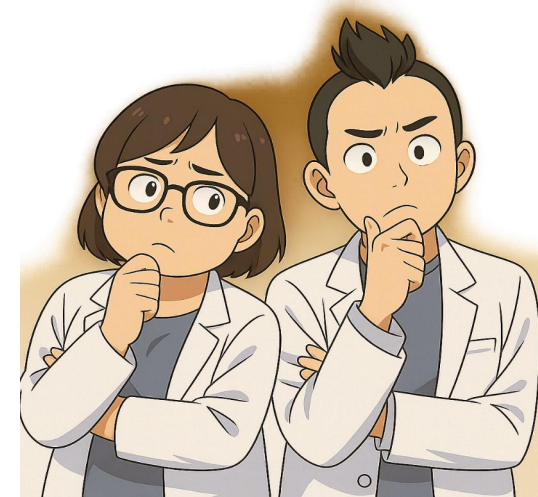
r's size gives you information about the **strength of the relationship**:



<https://www.nagwa.com/en/explainers/143190760373/>

Pearson correlation: p-value

How can I know that the relationship between the two variables are not just coincidence?



The **p-value** tells you how likely it is to observe a correlation as big as the one you found, if in reality there is no relationship between X and Y.

$p < 0.05$ means: There is less than a 5% chance that the correlation is just random.

Example sentence for reporting:

There was a significant positive correlation between study hours and exam scores, $r(248) = .45$, $p < .001$, indicating that students who studied more tended to score higher.

Pearson's r: $r(df) = \text{value}$, $p = \dots$

Your turn!

1. Copy folder Materials from to your local machine
 - <https://github.com/Data-Science-Center-UB/Intro-Quantitative-Analysis-R>
 - Path do subfolder Exercises will be your **working directory**
 - Folder content: data sets, exercises without & with solution
2. Do exercise “**03_qa_bi_ex**”
Version with solution: **03_qa_bi_ex_solution**



Multivariate Analyses

The background of the slide is a dark, deep blue. On the right side, there is a dense, fan-like pattern of fine, light blue lines or particle trails that radiate outwards from the center towards the top right corner. At the bottom of the image, there is a soft, glowing greenish-yellow arc or wave that curves across the width of the slide, adding a sense of depth and movement to the overall composition.

Correlation coefficient vs. Simple linear regression

Correlation measures tell you the **strength and direction of an association**.

- It does not assume one variable predicts the other;
- Correlation **does not mean causation**.

Regression models how one variable (X) **predicts** another (Y).

It gives you a mathematical equation ($Y = a + bX$).

- It tells you the size of the effect (slope);
- It includes significance tests and uncertainty (SE, p-value, CI).

Linear regression models the relationship between two variables with a straight line.

Simple linear regression is the specific case of linear regression with exactly **one predictor** variable.

Linear regression: Assumptions

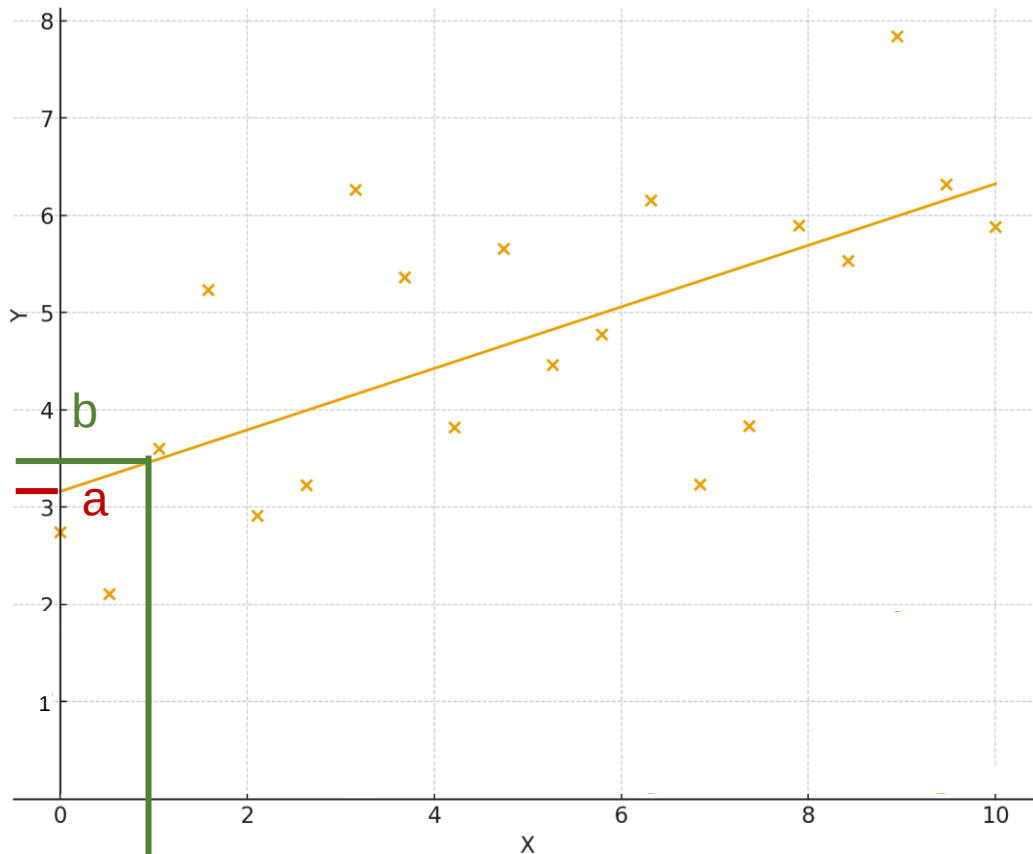
Linear regressions require:

- Dependent variable is **metric**.
→ Use logistic regression for categorical binary variables.
- Relationship between predictors and outcome is (approximately) **linear**.
- Residuals have **constant variance** (homoscedasticity).
- Residuals are approximately **normally distributed**.
- No strong **multicollinearity**, that means predictors are not (highly) redundant (e.g. age and years of education).

How does a simple linear regression look like?

Example: We model how **study time** predicts **exam performance**

$$(Y = a + bX).$$



X = Hours of study per week, Y = Exam score

Y = Dependent variable = exam score.

→ Variable you want to explain.

X = Predictor = hours studied.

→ Variable you assume has an effect on Y .

a = Intercept = exam score if hours studied = 0.

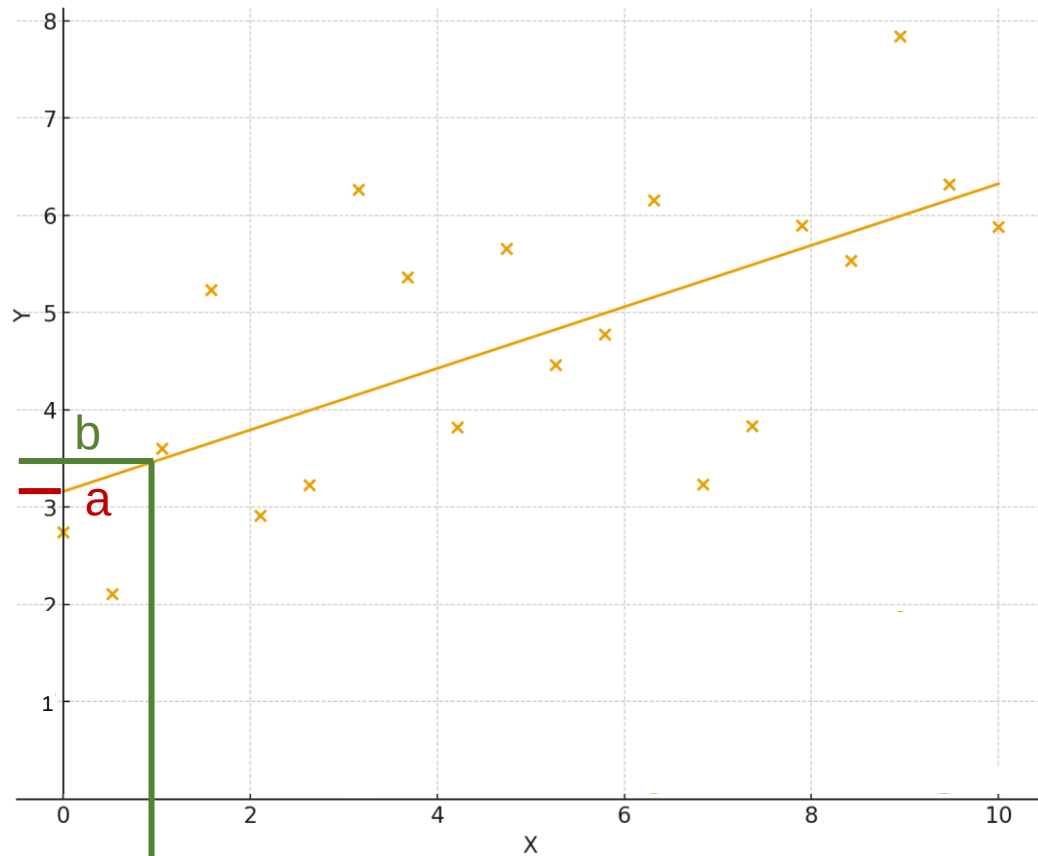
b = Slope (Steigung) = Increase in exam score per hour studied.

For each 1 unit increase in X , Y increases/decreases on average by b units.

- Positive b = increase, **positive effect on Y** .
- Negative b = decrease, **negative effect on Y** .

How does a simple linear regression look like?

Example: We model how **study time** predicts **exam performance**



X = Hours of study per week, Y = Exam score

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	3.162	0.503	—	6.280	0.000
h studied	0.316	0.086	0.655	3.674	0.002
R ²	0.429	—	—	—	—
n	20				

$$(Y = 3.162 + 0.316X).$$

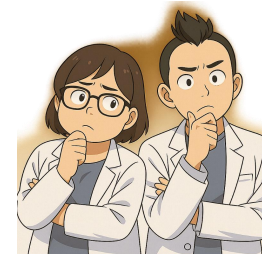
Without studying, the predicted exam score is 3.16.

For each 1h increase in hours studied, the exam score increases by 0.32.

What do the other outcomes tell me?

Example: We model how **study time** predicts **exam performance**

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	3.162	0.503	—	6.280	0.000
h studied	0.316	0.086	0.655	3.674	0.002
R ²	0.429	—	—	—	—
N	20				



Is this effect true or random?

P-value is the **probability of getting a result as extreme as (or more extreme than) the observed one**, if the null hypothesis were true (e.g. true effect = 0).

→ Small p-value (**< .05**) → rejects H_0 , **significant effect**.

What do the other outcomes tell me?

Example: We model how **study time** predicts **exam performance**

Term	Estimate	Std. Error	Beta	t	p-value	95% CI
Intercept	3.162	0.503	—	6.280	0.000	[2.10; 4.22]
h studied	0.316	0.086	0.655	3.674	0.002	[0.14; 0.50]
R ²	0.429	—	—	—	—	
N	20					

- The **standard error** reflects the uncertainty of b).
→ Smaller SE: more precise estimate.
- **Beta** is the regression coefficient in standard deviation units.
→ It shows how many SDs Y changes when X increases by one SD.
→ Useful to compare the strength of predictors measured on different scales.
- The **t value** is the estimate divided by its standard error.
→ Larger t: stronger evidence that true effect is not 0 (H₀ is rejected).
- From the estimate and its SE, we can build a **95% confidence interval**.
→ If **0 is not inside the 95% CI**, the effect is statistically **significant** at $\alpha = .05$.

What do the other outcomes tell me?

Example: We model how **study time** predicts **exam performance**

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	3.162	0.503	—	6.280	0.000
h studied	0.316	0.086	0.655	3.674	0.002
R ²	0.429	—	—	—	—
N	20				

Example:

“The model explains 43% of the variance in exam scores.”

R² shows **how much of the variance in Y is explained by the model**.

→ It is the proportion of variance in the outcome that can be predicted from the predictor(s).

R² ranges from **0 to 1**:

→ 0 = the model explains **none of the variance**.

→ 1 = the model explains **all of the variance**.

What is it?

Multivariate statistics deal with the analysis of relationships among **three or more variables at the same time**.

Why multivariate analyses are important

- **Understand complex relationships**, e.g. how an outcome (income) depends on several predictors at once (education, work experience, gender).
- **Control for confounders** and estimate the *unique* effect of one variable while holding others constant.
- **Improve prediction**, because combining several variables usually predicts outcomes better than a single predictor.

What decisions to make?

1. 🎯 Which variables to include in the model?

Depends on your **research question**, for example:

- What is your **outcome variable**? (e.g., income, satisfaction, voting intention)
- Which variables are your **main predictors**?
- Which variables should you **control for**? (e.g., age, gender, education)
- Are there **potential confounders** that might distort the relationship?
- Do you expect **interaction effects**? (e.g., education × gender)

2. ⚙️ Which type of analysis to use?

Depends mainly on the **level of measurement of the outcome variable**

Outcome Variable Y	Suitable analysis
Metric	Multiple linear regression
Categorical binary	Logistic regression

How does a multiple linear regression look like?

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	1512.438	552.317	—	2.738	0.007
training (ref. no)	187.532	228.917	0.058	0.819	0.414
Education: medium (ref. low)	214.759	297.531	0.067	0.722	0.472
Education: high (ref. low)	823.915	321.774	0.243	2.561	0.012
Gender: female (ref. male)	-573.284	219.847	-0.232	-2.608	0.010
Gender: diverse (ref. male)	-917.641	285.392	-0.291	-3.214	0.002
Age in years	41.763	12.311	0.272	3.393	0.001
R ²	0.253	—	—	—	—
Adjusted-R ²	0.223	—	—	—	—
n	150	—	—	—	—

Example: Linear regression on monthly gross income in €

What does the table tell us?

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	1512.438	552.317	—	2.738	0.007
Intercept (1512.44) <i>When training = 0, age = 0, gender = male and education = low, the expected income is about 1512 €.</i> (In practice, age = 0 is not meaningful, so the intercept is mostly a technical baseline.)					
Gender: female (ref. male)	-573.284	219.847	-0.232	-2.608	0.010
Gender: diverse (ref. male)	-917.641	285.392	-0.291	-3.214	0.002
Age in years	41.763	12.311	0.272	3.393	0.001
R ²	0.253	—	—	—	—
Adjusted-R ²	0.223	—	—	—	—
n	150	—	—	—	—

Example: Linear regression on monthly gross income in €

What does the table tell us?

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	1512.438	552.317	—	2.738	0.007
training (ref. no)	187.532	228.917	0.058	0.819	0.414
Training (187.53, n.s.) <i>Participants who completed the training earn on average about 188 € more than those without training, but this difference is <u>not statistically significant</u> when controlling for age, gender and education.</i>					
Gender: diverse (ref. male)	-917.641	285.392	-0.291	-3.214	0.002
Age in years	41.763	12.311	0.272	3.393	0.001
R ²	0.253	—	—	—	—
Adjusted-R ²	0.223	—	—	—	—
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Gender: diverse (ref. male)	-917.641	285.392	-0.291	-3.214	0.002

Gender: female (–573.28), diverse (–917.64) vs. male

- *Women earn on average about 573 € less than men, controlling for training, age and education.*
- *Persons in the “diverse” category earn on average about 918 € less than men, holding training, age and education constant.*

Example: Linear regression on monthly gross income in €

What does the table tell us?

Term	Estimate	Std. Error	Beta	t	p-value
Intercept	1512.438	552.317	—	2.738	0.007
training (ref. no)	187.532	228.917	0.058	0.819	0.414
Education: medium (ref. low)	214.759	297.531	0.067	0.722	0.472
Education: high (ref. low)	823.915	321.774	0.243	2.561	0.012

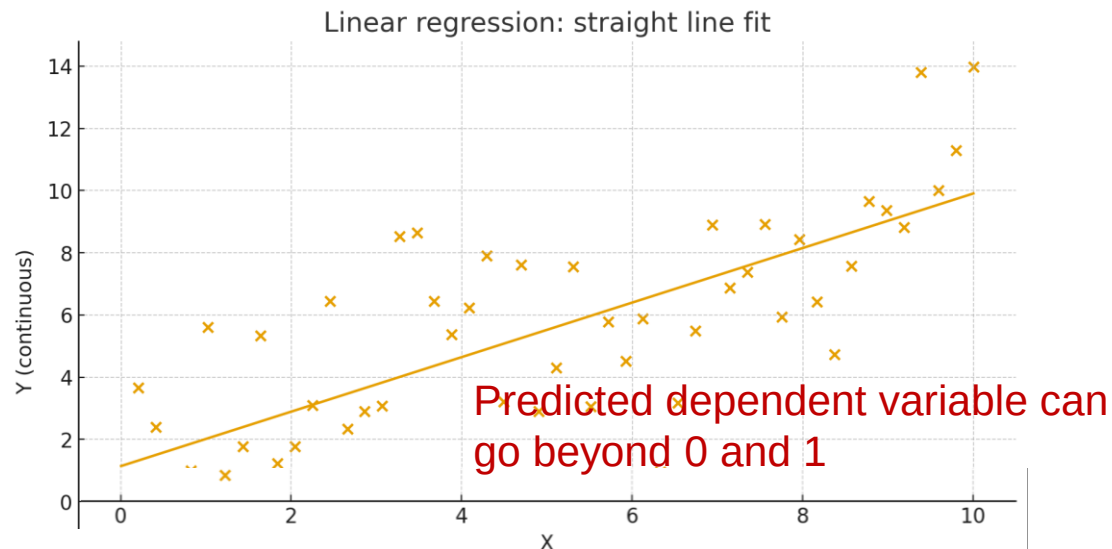
Age (41.76)

Holding training, education and gender constant, each additional year of age is associated with about 42 € higher monthly income.

Age in years	41.763	12.311	0.272	3.393	0.001
R ²	0.253	—	—	—	—
Adjusted-R ²	0.223	—	—	—	—
n	150	—	—	—	—

Example: Linear regression on monthly gross income in €

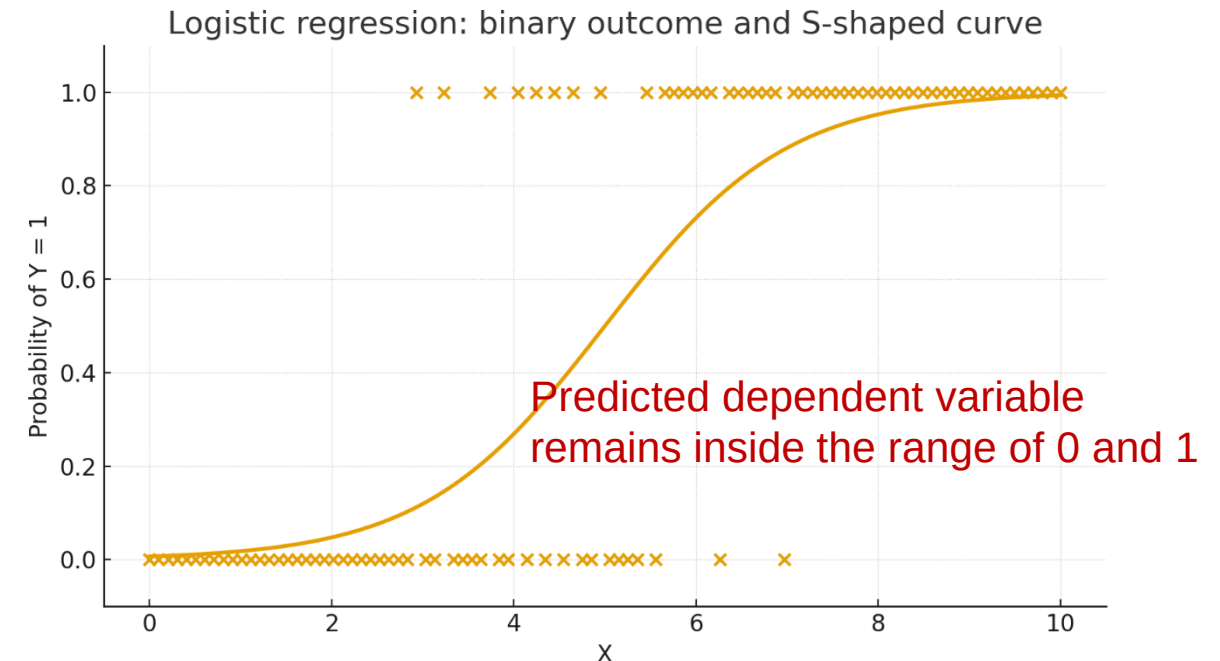
Linear vs logistic regression



Linear regression

Outcome **Y** is **metric/continuous** (e.g. income, test score).

Models a **straight-line relationship** between X and Y.



Logistic regression

Outcome **Y** is **categorical binary** (0/1, e.g. PhD started: yes/no).

Models a **curved, S-shaped relationship** between X and the **probability** of $Y = 1$.

How does a logistic regression look like?

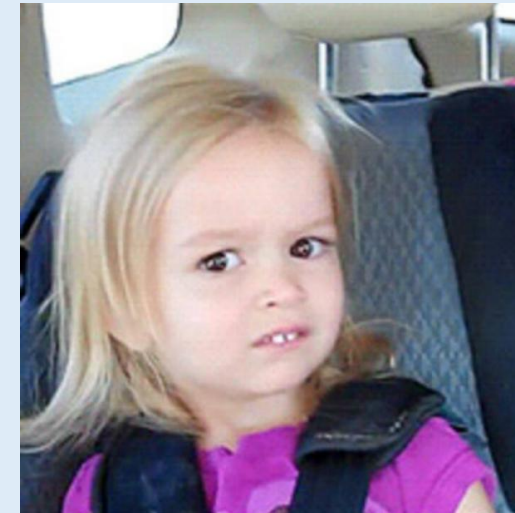
Term	Estimate	Std. Error	z	p-value
Intercept	-2.00	0.80	-2.50	0.012

Estimate for a predictor

How much the **log-odds** of $Y = 1$ change, if X_j increases by 1 unit, holding other variables constant.

Odds of an event are $odds = \frac{p}{1-p}$,
where p is the probability that the event happens.

Log-odds are the natural logarithm of the odds.



Age at graduation in years	-0.047	0.021	-2.26	0.024
Pseudo-R ²	0.18	—	—	—
n	450	—	—	—

Example: Logistic regression on PhD started (0 = no, 1 = yes)

Linear regression: Odds Ratios

Odds ratio (OR) compares how likely an event is in one group (or at one value of a predictor) to how likely it is in another.

$$OR = \frac{\text{Odds in group 1}}{\text{Odds in group 2}}$$

For **categorical predictors**, group 1 is the category you are estimating the effect for (e.g. “yes”, “female”), and group 2 is the reference category. The odds ratio compares the odds in group 1 to the odds in group 2.

For **continuous predictors**, the odds ratio compares the odds at $X + 1$ unit (group 1) to the odds at X (group 2), e.g. With each additional year of age, the odds of starting a PhD are multiplied by 0.95 (i.e. they decrease by about 5%).

$OR = 1 \rightarrow$ **no difference** in odds between the groups.

$OR > 1 \rightarrow$ **higher odds** in the first group (event is more likely).

$OR < 1 \rightarrow$ **lower odds** in the first group (event is less likely).

What does the table tell us?

Term	OR	Std. Error	z	p-value
Intercept	0.06	0.80	-3.63	0.000
Worked as RA (ref. no)	2.51	0.27	3.41	0.001

Worked as RA (OR = 2.51)

*Graduates who worked as a research assistant (RA) have about **2.5 times higher odds** of starting a PhD than graduates who did not work as an RA, **controlling for discipline, final grade, gender and age.***

*→ RA experience is associated with a **significant higher chance** of starting a PhD.*

Gender: female (ref. male)	0.71	0.24	-1.46	0.144
Gender: diverse (ref. male)	0.33	0.40	-2.75	0.006
Age at graduation in years	0.95	0.021	-2.26	0.024
Pseudo-R ²	0.18	—	—	—
n	450	—	—	—

Example: Logistic regression on PhD started (0 = no, 1 = yes)

What does the table tell us?

Term	OR	Std. Error	z	p-value
Intercept	0.06	0.80	-3.63	0.000
Worked as RA (ref. no)	2.51	0.27	3.41	0.001
Final grade (“very good” vs “worse”)	2.34	0.25	3.40	0.001
Discipline: social sciences (ref. nat.)	0.74	0.29	-1.03	0.302
Discipline: humanities (ref. nat.)	0.43	0.33	-2.58	0.000
Discipline: health sciences (ref. nat)	1.16	0.34	0.44	0.662

Social sciences (OR = 0.74), Humanities (OR = 0.43)

Graduates from the **social sciences** have about **0.74 times the odds** of starting a PhD compared to graduates from the **natural sciences**, controlling for RA experience, grade, gender and age (this difference is not statistically significant).

Graduates from the **humanities** have only about **0.43 times the odds** of starting a PhD compared to **natural sciences** graduates – that is, their odds are clearly lower, even after controlling for RA experience, grade, gender and age....

Example: Logistic regression on PhD started (0 = no, 1 = yes)

What does the table tell us?

Term	OR	Std. Error	z	p-value
Intercept	0.06	0.80	-3.63	0.000
Worked as RA (ref. no)	2.51	0.27	3.41	0.001
Final grade (“very good” vs “worse”)	2.34	0.25	3.40	0.001
Discipline: social sciences (ref. nat.)	0.74	0.29	-1.03	0.302
Discipline: humanities (ref. nat.)	0.43	0.33	-2.58	0.000

Age at graduation (OR = 0.95)

*For each additional year of age at graduation, the odds of starting a PhD are multiplied by about **0.95**, that is, they **decrease by roughly 5% per year**, controlling for discipline, RA experience, grade and gender.*

Age at graduation in years	0.95	0.021	-2.26	0.024
Pseudo-R ²	0.18	—	—	—
n	450	—	—	—

Example: Logistic regression on PhD started (0 = no, 1 = yes)

Pseudo- R^2

In **logistic regression**, the outcome is **binary** and we do not have the same variance decomposition.

- Instead, we use **pseudo R^2 measures** (e.g. McFadden, Cox–Snell, Nagelkerke) that compare the fitted model to a **null model** (with no predictors) based on **likelihoods**.
- They show **how much better** the model fits compared to a model without predictors, **not** the exact “percentage of variance explained”.
- Pseudo R^2 values are usually **much smaller** than R^2 in linear regression (e.g. 0.10–0.30 can already indicate a useful model).

What does the table tell us?

Term	OR	Std. Error	z	p-value
Intercept	0.06	0.80	-3.63	0.000
Worked as RA (ref. no)	2.51	0.27	3.41	0.001
Final grade (“very good” vs “worse”)	2.34	0.25	3.40	0.001
Discipline: social sciences (ref. nat.)	0.74	0.29	-1.03	0.302
Discipline: humanities (ref. nat.)	0.43	0.33	-2.58	0.000

Pseudo-R²

The pseudo-R² of 0.18 suggests that the logistic regression model provides a modest improvement in explaining who starts a PhD compared to a model without predictors.

Age at graduation in years	0.95	0.021	-2.26	0.024
Pseudo-R ²	0.18	—	—	—
n	450	—	—	—

Example: Logistic regression on PhD started (0 = no, 1 = yes)

Your turn!

1. Copy folder Materials from to your local machine
 - <https://github.com/Data-Science-Center-UB/Intro-Quantitative-Analysis-R>
 - Path do subfolder Exercises will be your **working directory**
 - Folder content: data sets, exercises without & with solution
2. Do exercise “**04_qa_multi_ex**”
Version with solution: **04_qa_multi_ex_solution**



The background of the slide is a dark blue gradient. On the right side, there is a dense, fan-like pattern of bright blue lines and dots, resembling a particle trail or a data visualization. The lines radiate from the bottom right towards the top right. The text "Wrap-up" is centered in the middle of the slide.

Wrap-up

Resources



DataNord



University of Bremen

Self-Learning Tool

<https://datasciencebox.org/>

Cheat Sheets

<https://posit.co/resources/cheatsheets/>

The screenshot shows the R4DS course content page. It lists units from Unit 1 to Unit 10. Unit 2, 'Deck 5: Tidy data', is highlighted. It includes links for Slides, Source, and Video. The 'Reading' section for Unit 2 is 'JSS: Tidy data'. Unit 6, 'Grammar of data wrangling', and Unit 7, 'Working with a single data frame', are also visible.

The screenshot shows a comprehensive cheat sheet for dplyr functions. It is organized into sections: 'Summarize Cases', 'Group Cases', 'Manipulate Cases', 'Manipulate Variables', 'Logical and boolean operators to use with filter()', 'ARRANGE CASES', and 'ADD CASES'. Each section contains a list of functions with brief descriptions and code examples. The cheat sheet is attributed to 'posit' and includes a copyright notice for 2025-08.

Literature

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DataNord Trainings & Workshops

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... more on dsc-ub.de/en/qualification

CURRENT TRAININGS



MARCH

11 No Fear of Numbers – Introduction to Quantitative Data Analysis in R
09:00 - 16:30 | Campus

17-18 Introduction to Python for Researchers
09:30 - 16:00 | Campus

APRIL

08-09 Applied Text Mining Using Python
09:30 - 17:00 | Campus

14 Create your own Data Management Plan: A Hands-on Workshop for Quantitative Data
09:00 - 13:00 | Campus

21 Responsible Use of Personal and Sensitive Data in Research
09:30 - 16:00 | Campus

22 Getting Git-Ready for Research Projects
09:00 - 16:30 | Campus

MAY

19-20 Geospatial Data Visualization and Maps in Python
09:30 - 16:00 | Campus

JUNE

04 From Audio to Text: Automated Transcriptions with Whisper
09:30 - 16:30 | Campus

Additional workshops will be announced on website.

DataNord Consulting Service

How can I create a data management plan?

How do I prepare my data to apply multivariate statistics?

What do I need to consider when collecting personal data?



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- Support in the application phase and research process
- Free of charge!
- More information: <https://dsc-ub.de/en/consultation.php>



In the space below please write any overall comments about this course or instructor not covered above.

If I had one hour to live,
I'd spend it in this
class because it feels
like an eternity.

<https://www.dsc-ub.de/evaluation-DSC-2026-01.php>



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Thank You for Your Participation!

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